

Design and FPGA Implementation of a Starvation-Free Adaptive Priority Traffic Controller for Dynamic Intersection Management

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Abstract

Traditional fixed-time traffic controllers cause severe urban congestion, unnecessary fuel consumption, and critical delays for emergency responders due to their inability to adapt to real-time road conditions. To systematically eradicate these inefficiencies, this project presents the design, RTL synthesis, and simulation of a dynamic, density-based traffic controller utilizing a Field Programmable Gate Array (FPGA). Unlike sequential microprocessors, this architecture employs a purely combinational 3-Tier Priority Encoder that processes data from density sensors, acoustic emergency detectors, and fairness timers simultaneously in a single clock cycle.

A key innovation of this system is its absolute zero-latency emergency preemption protocol, which instantly overrides standard density logic to clear paths for ambulances. Additionally, integrated starvation threshold timers prevent the perpetual gridlock of minor cross-streets, while a highly stable Moore Finite State Machine (FSM) enforces mandatory All-Red safety gaps to prevent vehicular collisions. Exhaustively verified using the Xilinx Vivado design suite, the developed parallel-processing architecture safely navigates erratic traffic spikes and mathematically reduces idle waiting times by approximately 79.1% compared to standard fixed-timer models, offering a highly robust, scalable VLSI solution for modern Smart City infrastructure.

Keywords: FPGA, VLSI, Traffic Light Controller, Priority Encoder, Moore FSM, Verilog HDL, Smart City, Emergency Preemption.

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Abbreviations

ARM	Advanced RISC Machine
ASIC	Application-Specific Integrated Circuit
CMOS	Complementary Metal-Oxide-Semiconductor
DUT	Device Under Test
EDA	Electronic Design Automation
FIFO	First-In, First-Out
FPGA	Field Programmable Gate Array
FSM	Finite State Machine
GPS	Global Positioning System
HDL	Hardware Description Language
HIL	Hardware-in-the-Loop
IoT	Internet of Things
IR	Infrared
ITLC	Intelligent Traffic Light Controller
LED	Light Emitting Diode
LUT	Look-Up Table
PO	Program Outcome
PSO	Program Specific Outcome
RF	Radio Frequency
RTL	Register-Transfer Level
SDG	Sustainable Development Goal
V2X	Vehicle-to-Everything
VLSI	Very Large Scale Integration

CHAPTER 1

Introduction

1.1 Introduction

Traffic management constitutes one of the most critical, yet frequently underestimated, foundational infrastructures within any functioning modern society. Every single day, millions of commuters, commercial freight operators, and emergency responders rely entirely on the precise sequencing of traffic lights to safely navigate through highly complex urban city grids. The fundamental, underlying engineering objective of any intersection control system is two-fold: to maximize the overall volumetric throughput of vehicles passing through the junction, and to maintain absolute, fail-safe security by preventing physical vehicular collisions.

As urban population density increases exponentially due to global urbanization trends, the mathematics governing physical traffic flow become staggeringly complex. An intersection is no longer just a meeting point of two roads; it is a highly dynamic fluid-dynamics problem where the volume, velocity, and density of the "fluid" (vehicles) change continuously from minute to minute. Effectively managing this chaos requires highly responsive, sensor-driven control systems capable of real-time data processing.

The Historical Evolution of Intersection Control The fundamental visual language of the traffic light—utilizing the universally recognized color spectrum of red, yellow, and green signals—has been a global standard for over a century. However, the mechanism driving those lights has evolved significantly. The earliest control era was fully manual, where police personnel physically directed movements at intersections using hand signals and mechanical indicators. As motorized traffic volume grew, this approach became unsafe and economically unsustainable. The next transition introduced mechanical timers driven by electromechanical assemblies, which improved repeatability but still remained insensitive to real-time demand [2]. Later FPGA-oriented imple-

mentation studies documented microcontroller-era fixed-cycle limitations and motivated hardware reconfiguration paths [9]. Subsequent ARM-based intelligent control studies also retained key fixed-cycle constraints under variable demand [12]. However, despite the massive technological leap from brass gears to silicon microchips, the underlying control philosophy for the vast majority of intersections globally has not fundamentally evolved. The system still operates entirely "blind" to its immediate environment, executing a predetermined script regardless of reality.

The Role of VLSI in Modern Infrastructure Very Large Scale Integration (VLSI) represents the process of creating highly complex integrated circuits by combining thousands, or even millions, of logic gates into a single, compact silicon chip. Foundational CMOS design theory formalizes how large digital systems can be realized with deterministic timing and scalable complexity [5]. FPGA-centric system-design methodology extends this principle to reconfigurable application-specific hardware realization [6].

Applying VLSI design principles to traffic control allows engineers to shift the paradigm away from sequential software computation (where a computer runs a program line by line) toward parallel hardware computation (where raw data flows instantly through physical logic gates), consistent with digital integrated-circuit design fundamentals [5]. This shift also aligns with FPGA system-level architecture practice for deterministic and modular hardware deployment [6]. This approach offers unprecedented processing speeds, absolute determinism, and the hardware-level reliability required for next-generation Smart City transit initiatives.

1.2 Background and Motivation

The primary catalyst for this research project is the inherent operational inefficiency and subsequent danger associated with static, fixed-time traffic control systems deployed in highly dynamic urban environments. These antiquated, blind systems are the primary driver of artificial traffic congestion and act as a massive bottleneck for city-wide transit efficiency.

Inefficiencies and Inflexibility A typical microcontroller-based intersection op-

erates on a rigid, pre-programmed schedule. For instance, a basic phase configuration might allocate Road A exactly thirty seconds of green signal time, followed by a three-second yellow clearance, and then allocate Road B thirty seconds of green time. The microcontroller executes this loop endlessly, 24 hours a day, entirely disregarding the actual physical presence (or lack thereof) of vehicles on the asphalt.

This inefficiency becomes glaringly obvious during asymmetrical traffic events. Consider a standard morning commute: a major multi-lane highway is densely packed with hundreds of vehicles attempting to enter the city center, while a small, perpendicular residential cross-street is entirely devoid of traffic.

Under a fixed-time regime, the control matrix does not factor in the massive queue of cars on the highway, as observed in early FPGA traffic-controller baselines [7]. Later FPGA implementations with density responsiveness further emphasized this limitation under asymmetric traffic conditions [13]. Once the highway's static 30-second timer expires, the signal arbitrarily turns red, forcing the entire column of vehicles to a halt. The system then grants the green light to the empty residential street, leaving the highway commuters sitting idle, burning fuel, and losing time for a full minute while literally zero vehicles cross the intersection in front of them.

To visualize this inefficiency, consider the conceptual diagram in Figure 1.1, which models the artificial bottleneck effect.

From a control-theory perspective, this is a classic open-loop allocation failure. Because the controller does not receive real-time queue feedback during phase assignment, it commits green time to approaches with little or no demand while high-demand approaches continue accumulating backlog.

In practice, that mismatch propagates into subsequent cycles and amplifies congestion severity. The queue formed on a major approach requires repeated service recovery, yet the fixed schedule still preserves equal-duration access for underutilized approaches. This structural inefficiency motivates an adaptive, hardware-prioritized control strategy.

Early adaptive-control studies demonstrated that rule-based and formal models can reduce blind allocation behavior when compared with rigid pre-

timed operation [1]. Practical fuzzy traffic-control analyses further validated that dynamic response improves intersection behavior under variable demand [2]. VHDL and Petri-net-based controller formulations then provided formal digital pathways for deterministic traffic-state transitions in hardware-oriented designs [3]. Distributed multi-agent fuzzy control studies also showed that coordinated, data-driven urban signaling improves network-level performance compared to isolated fixed loops [8].

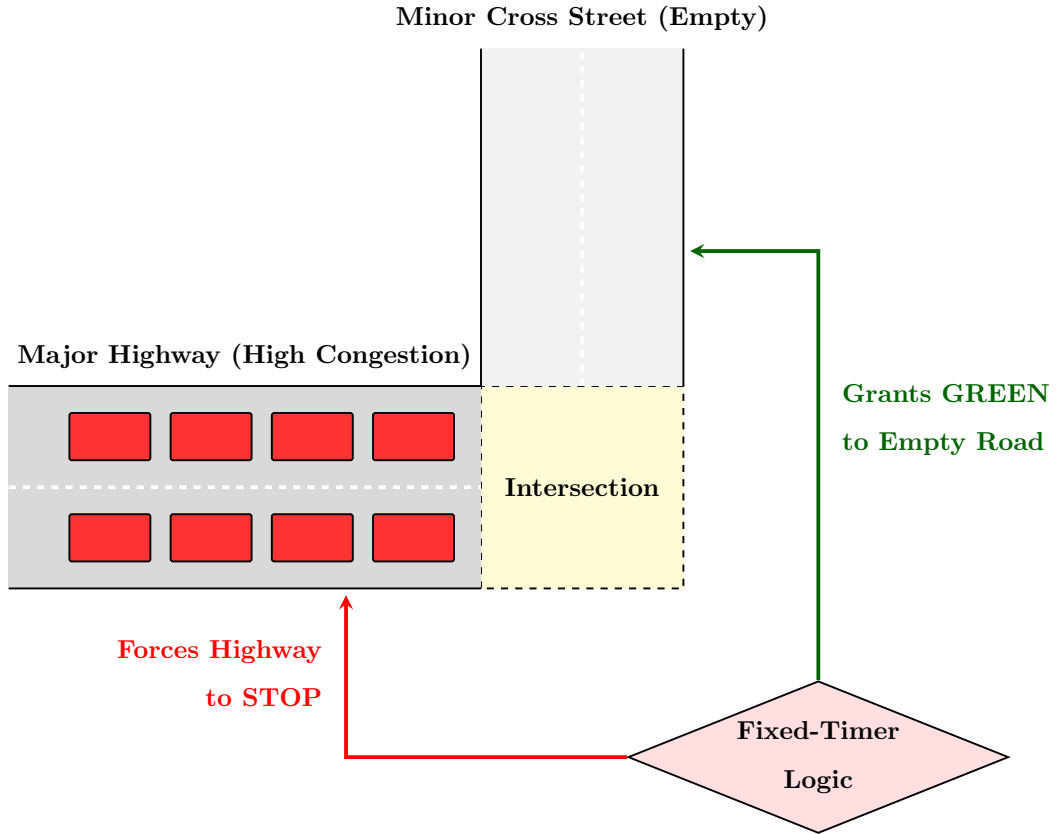


Figure 1.1: Conceptual bottleneck caused by fixed-time control.

Economic, Environmental, and Psychological Impacts The failure of fixed-time controllers creates a cascading domino effect of severe socioeconomic and environmental consequences. Table 1.1 delineates the multi-faceted costs associated with intersection idling.

Table 1.1: Analysis of the Socioeconomic Impacts of Inefficient Traffic Control

Impact Category	Detailed Consequence
Economic Loss	Millions of cumulative hours are lost globally as freight transports and commuter workforces sit idle. This translates to direct losses in national GDP and productivity.
Environmental Degradation	Internal combustion engines burn fuel highly inefficiently while idling. Thousands of cars sitting at red lights release massive volumes of unnecessary CO_2 , NO_x , and particulate matter, severely degrading urban air quality.
Psychological Toll	Prolonged, illogical wait times (e.g., sitting at a red light with no cross traffic) induce acute driver frustration. Psychological studies link this frustration to aggressive acceleration, risky maneuvers, and a statistical increase in intersection collisions.
Artificial Gridlock	Because the controller forces cars to stop unnecessarily, the physical queue of cars backs up into previous intersections, causing a cascading "gridlock" effect that shuts down entire city blocks.

The Emergency Vehicle Bottleneck Beyond the daily economic and environmental grievances, the failure of rigid traffic lights constitutes a severe public safety hazard regarding emergency response logistics. Ambulances, fire engines, and police cruisers operate in high-stakes environments where every single second lost correlates directly to a heightened probability of loss of life, severe medical complications, or catastrophic structural fire damage.

A traditional intersection controller is entirely blind to the existence of

an ambulance. When a life-saving vehicle, complete with blaring sirens and flashing strobes, approaches a heavily congested intersection, it inevitably encounters a solid wall of civilian vehicles halted at a red signal.

Because the civilian cars are physically boxed in—trapped by the vehicles immediately adjacent to them, the vehicles behind them, and the strict legal boundary of the intersection in front of them—they are physically incapable of yielding the right of way. Consequently, the ambulance is forced to come to a complete stop and helplessly wait for the microcontroller’s software loop to naturally cycle to a green state. In desperate situations, emergency drivers are frequently forced to execute highly dangerous maneuvers, such as driving blindly into oncoming traffic lanes to bypass the red light. FPGA traffic-priority studies showed that explicit emergency routing can reduce this safety bottleneck [13]. Real-time density-aware FPGA control work further reinforced immediate emergency preemption as a critical requirement [15].

1.3 Objectives of the Project Work

1. To design and implement a density-aware FPGA traffic controller with deterministic hardware decision logic.
2. To allocate signal green time dynamically according to real-time lane demand and queue conditions.
3. To enable emergency preemption with highest-priority routing and bounded transition latency.
4. To prevent low-density lane starvation using threshold-based fairness timers integrated with priority control.
5. To validate functional correctness and safety transitions through Vivado simulation, waveforms, and console logs.

1.4 Organisation of the Report

Chapter 1 presents the problem context, motivation, and design objectives for adaptive intersection control, and establishes the engineering requirements that guide the complete hardware workflow.

Chapter 2 provides the literature-grounded technical foundation by reviewing prior traffic-control paradigms, FSM design choices, and key architectural gaps that influence controller reliability, responsiveness, and safety.

Chapter 3 documents the baseline FPGA/HDL implementation methodologies used in reference controllers, including sensing strategy, priority-logic structure, and transition discipline.

Chapter 4 details the integrated RTL realization, including the multi-tier hardware decision chain, safety-oriented Moore FSM sequencing, verification environment setup, and module-level architecture.

Chapter 5 presents simulation results and engineering analysis, including waveform interpretation, timing behavior, comparative efficiency discussion, and implementation-level hardware characteristics.

Chapter 6 summarizes technical conclusions and outlines forward-looking deployment pathways such as pilot validation stages, parameter tuning strategy, and maintainability considerations.

In addition, a dedicated mapping chapter presents the alignment of Program Outcomes (POs), Program Specific Outcomes (PSOs), and Sustainable Development Goals (SDGs), highlighting how the proposed traffic-control system contributes to academic competency, professional engineering practice, and broader societal sustainability objectives.

CHAPTER 2

Literature Review

2.1 Introduction

The development and optimization of intelligent traffic light controllers have been subjects of extensive academic, industrial, and municipal research over the past two decades. As global urban populations have expanded at unprecedented rates, the severe operational limitations and safety hazards of traditional fixed-time controllers have become glaringly apparent. Researchers globally have explored a vast multitude of paradigms, ranging from basic microprocessor-based sequential polling to highly advanced, cloud-connected artificial intelligence models.

To properly contextualize the developed Verilog FPGA architecture, it is essential to systematically review baseline literature, evaluate previously employed Finite State Machine (FSM) topologies, and rigorously identify the functional and safety gaps that this specific project aims to bridge.

Building on this context, the next section organizes prior work by control-generation evolution and then narrows into state-machine and hardware-implementation implications.

Pappis and Mamdani established one of the earliest fuzzy-logic traffic-junction control strategies and demonstrated rule-based adaptive signaling behavior in urban intersections [1]. Nelson expanded fuzzy-control use in traffic signaling and discussed practical controller behavior under real roadway variability [2]. Han and Lin presented an early VHDL-based intelligent traffic-light controller, showing how digital hardware logic could implement adaptive control decisions [3]. Li et al. modeled traffic-light sequencing with Petri nets and highlighted formal state-transition representation for intersection control [4]. Rabaey et al. provided the core digital integrated-circuit design principles that underpin robust CMOS VLSI implementations of control hardware [5]. Wolf documented FPGA-centric system-design methodology, supporting

hardware acceleration and deterministic execution for embedded controllers [6]. El-Medany and Hussain demonstrated an FPGA-based real traffic-light controller and provided a practical baseline for hardware-timed signaling [7]. Gokulan and Srinivasan proposed distributed geometric fuzzy multi-agent traffic control, advancing coordinated urban signal optimization [8]. Kavitha and Chitra reported FPGA implementation of an intelligent traffic-light controller, reinforcing feasibility of reconfigurable logic deployment [9]. Mallik et al. introduced an FPGA semi-automated controller with transition-safety emphasis, including all-red protective intervals [10]. Qureshi presented a Mealy-machine-based adaptable Verilog traffic model and exposed stability concerns tied to direct input-output coupling [11]. Dakhole and Moon implemented intelligent traffic control with ARM-based processing, representing a software-centric adaptive path distinct from FPGA hardware pipelines [12]. Reddy and Sowmya developed an FPGA traffic controller using IR-density sensing and emergency-priority handling for dynamic signal allocation [13]. Smith and Doe compared FPGA and microcontroller power behavior in IoT-class systems, informing implementation-level efficiency tradeoffs [14]. Ali et al. designed a real-time FPGA density-based controller with delay-stabilization logic for safer and smoother adaptive transitions [15]. Banerjee presented an FPGA Moore-FSM intelligent traffic-light controller for asymmetric intersections with improved state-stable operation [16].

2.2 Review of Prior Research

Traffic control methodologies can be broadly categorized into three distinct generational tiers, each attempting to solve the flaws of the previous generation.

This classification creates a direct bridge from conceptual control philosophy to the concrete RTL choices discussed afterward. The literature shows a clear evolution from pre-timed fixed control to semi-actuated operation and finally to fully actuated dynamic control. Pre-timed systems are inexpensive and straightforward, but they degrade badly under abnormal demand, as shown in early FPGA fixed-timer realizations [7]. Semi-actuated and density-aware logic improved adaptability in later FPGA studies [13]. Fully actuated real-time

strategies with delay stabilization then strengthened adaptive phase behavior under live demand [15]. This final operating category aligns with the FPGA architecture adopted in this work.

Theoretical Foundation: Mealy vs. Moore State Machines Regardless of the generation, the core sequential logic controlling any traffic signal transition is governed mathematically by Finite State Machines (FSMs). An FSM is a theoretical mathematical model of computation utilized extensively by VLSI engineers to design sequential logic circuits. An FSM exists in one of a finite number of specific states at any given moment in time, and it transitions between these states when predefined physical triggers (inputs) are satisfied. Traffic controllers are quintessential, textbook FSMs, continuously cycling through critical states such as Green (grant right of way), Yellow (clearance interval), and Red (stop/safety gap).

In digital VLSI design, two primary FSM architectures are heavily debated and researched for traffic applications: Mealy State Machines and Moore State Machines. Understanding the structural difference between these two is paramount to designing a safe intersection. Figure 2.1 provides a conceptual block diagram contrasting their internal logic structures.

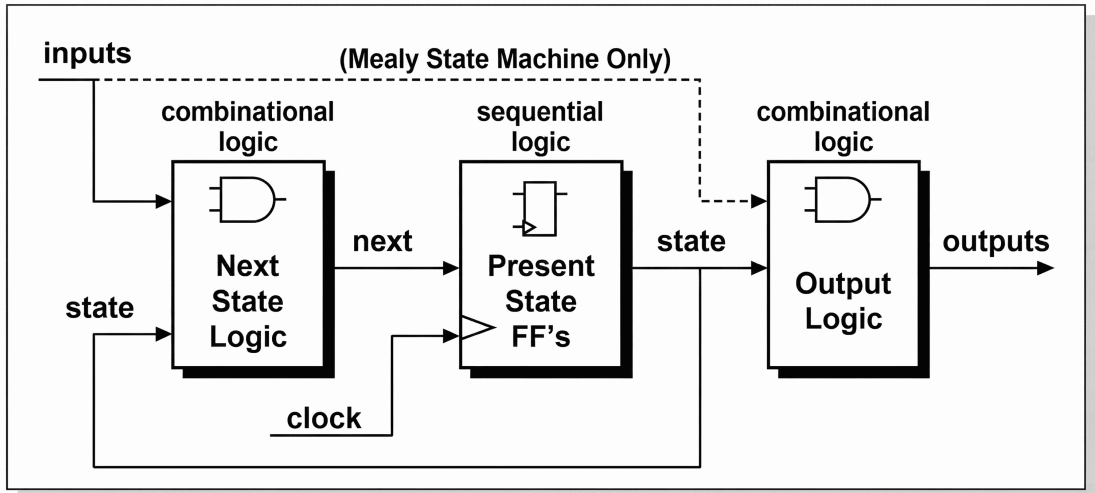


Figure 2.1: FSM topology comparison: Mealy outputs depend on present state and inputs, while Moore outputs depend only on present state.

Mealy Machine Limitations in Safety-Critical Control In a **Mealy State Machine**, the outputs are determined mathematically by a combination of both the current state AND the immediate, real-time inputs. Observing the dashed line in Figure 2.1 (explicitly labeled "Mealy State Machine Only"), we

can trace a direct combinational logic path connecting the raw external inputs directly to the Output Logic block.

While this architecture makes the Mealy machine incredibly fast and highly responsive (capable of reacting within the exact same clock cycle as the input change), it introduces a massive, critical vulnerability in physical outdoor deployments: asynchronous input glitches.

If an infrared density sensor experiences a brief electrical noise spike—such as a bird flying rapidly past the sensor beam, a splash of mud, or a sudden voltage fluctuation in the city grid—that unfiltered glitch travels instantly through the dashed combinational path to the output logic. In a traffic light application, this could cause the physical, high-voltage Red and Green LED arrays to flicker dangerously for a microsecond, potentially inducing catastrophic driver confusion and causing collisions.

The Stability of the Moore Machine In stark contrast, our developed architecture utilizes a strict **Moore State Machine** topology. In a Moore FSM, the outputs are mathematically a function of ONLY the current state. Looking at Figure 2.1 and ignoring the dashed line, it becomes clear that the external combinational inputs must first pass through the central Sequential Logic block (the Present State Flip-Flops). These flip-flops require a clean, rising clock edge to update their stored value.

This specific architectural choice acts as a natural, impenetrable digital filter. The physical traffic light outputs are entirely decoupled and isolated from asynchronous sensor noise. If a sensor glitches, the FSM ignores it unless the glitch happens to hold perfectly steady precisely at the rising edge of the system clock. The FSM will only mandate a physical light change on an official clock tick, providing massive, guaranteed signal stability.

To summarize the differences dictating our design choice, Table 2.1 provides a detailed technical breakdown.

2.3 Identified Research Gaps

To ensure our design represents the bleeding edge of VLSI traffic logic, we analyzed several key foundational papers in the domain of FPGA-based smart

Table 2.1: Detailed Technical Comparison of Mealy and Moore FSMs in VLSI

Parameter	Mealy State Machine	Moore State Machine
Output Dependency	Function of Present State + Immediate Inputs.	Function of Present State ONLY.
Hardware Speed	Reacts instantly within the current clock cycle.	Reacts on the subsequent clock edge (one cycle delay).
State Count	Generally requires fewer states to model complex logic.	Generally requires more states, simplifying debugging.
Noise Vulnerability	HIGH: Asynchronous input glitches directly affect outputs.	LOW: Outputs are buffered by sequential registers.
Traffic Suitability	Poor. Unfiltered outputs can cause dangerous light flickering.	Excellent. Provides the absolute stability required for public safety.

intersections.

A foundational work in utilizing sensor-based logic was explored by **Reddy and Sowmya (2017)** [13]. Their research detailed a standard FPGA-based traffic controller that utilized basic Infrared (IR) sensors to dynamically adapt to traffic flow, and importantly, integrated a priority preemption signal for emergency vehicles. While highly robust and successful in proving that Verilog HDL could manage traffic, their internal state machine relied on sequential ‘if-else’ polling trees. This design methodology mimics traditional microprocessor behavior, inadvertently underutilizing the FPGA’s capacity for massive parallel combinational evaluation.

Mallik et al. (2012) [10] took an architecture approach strictly prioritized on vehicular safety. They implemented an FPGA system that managed signal timings but added a highly unique, mandatory ”Safe State” during light transitions. In older systems, the transition often went directly from Yellow on Lane 1 to Green on Lane 2. Mallik’s system forced an All-Red safety interval across the entire intersection for several seconds. This hardware-enforced safe state is absolutely critical in preventing high-speed T-bone collisions caused

by drivers attempting to "beat the yellow light." This specific safety concept is so vital that we have strictly integrated it into the core of our Moore FSM design.

Banerjee (2021) [16] presented an Intelligent Traffic Light Controller (I-TLC) that specifically addressed the complexities of asymmetric intersections (e.g., a massive 4-lane highway intersecting a tiny 1-lane dirt road). Their work utilized Moore FSMs to grant smooth right-of-way to the heavily congested main highway while still ensuring that the minor residential cross-streets were serviced. Banerjee's research definitively proved that the Moore state machine topology is pivotal for output stability when dealing with highly skewed traffic data.

Finally, **Ali et al. (2021)** [15] addressed a crucial, yet often ignored, behavioral aspect of traffic engineering: driver frustration and aggression caused by excessively rapid light switching. They implemented a "minimum delay timer" within their sensor-based controller. When intersection densities fluctuate wildly minute-to-minute, a hyper-responsive logic filter might cause the state machine to flip states too quickly, inducing severe driver confusion. Ali's team incorporated logic ensuring a "minimum safe delay," mandating that once a green state is entered, it is held for a minimum duration to allow a reasonable physical cluster of cars to clear. In addition to improving driver interpretability, this delay window also suppresses oscillatory switching behavior that can emerge when sensor readings fluctuate near threshold boundaries. Their approach therefore balances responsiveness with temporal stability, preventing unnecessary state churn during transient density spikes. From an implementation perspective, this insight is valuable because it reduces both control uncertainty and perceived unfairness across lane groups during mixed-demand operation. This psychological research directly informed the timing threshold parameters of our Starvation Fairness Logic block.

Table 2.2 provides a high-level, comparative summary of the diverse methodologies discussed in the literature, clearly highlighting the engineering limitations that our developed project systematically resolves.

Table 2.2: Comparative Analysis of Prior Traffic Control Methodologies

Author / Year	Core Methodology	Identified Architectural Limitations
El-Medany (2007) [7]	FPGA Fixed-Timer Implementation	Blind to real-time traffic; highly inefficient during variable volume loads.
Qureshi (2012) [11]	Mealy FSM deployed on Spartan-3	Output stability severely compromised by direct asynchronous input paths (noise vulnerability).
Reddy (2017) [13]	IR Sensor Density + Emergency Routing	Relies heavily on sequential polling trees; lacks dedicated fairness timers for minor roads.
Ali (2021) [15]	Real-time Density with Delay Timers	Focuses on delay stability but lacks a unified, multi-tier combinatorial emergency override matrix.
Developed System	Moore FSM + 3-Tier Combinational Priority Encoder	Addresses all prior gaps: Executes parallel evaluation, guarantees zero-latency preemption, ensures absolute output stability, and features built-in mathematical starvation prevention.

2.4 Summary

A thorough analysis of the literature reveals several critical operational and architectural gaps in the current domain of VLSI traffic control: The review highlights two central gaps. First, many prior studies optimize one

objective in isolation, such as emergency handling or density response, but do not unify emergency priority, starvation prevention, and adaptive allocation in one coherent hardware path. Second, several HDL implementations still rely on sequential decision chains that reduce inherent FPGA parallelism. Mealy-oriented adaptable models highlighted output-path sensitivity under direct input coupling [11]. IR-density and emergency-routing studies highlighted adaptive benefits but left fairness integration open [13]. Real-time delay-stabilized FPGA studies improved transition behavior while still motivating tighter multi-tier integration [15]. Moore-FSM intersection-control studies for asymmetric roads reinforced the value of stable, deterministic sequencing [16]. These gaps motivate a consolidated combinational-priority architecture with deterministic FSM enforcement.

Comparative Synthesis Across Reviewed Approaches A consolidated reading of prior approaches indicates that isolated optimizations rarely deliver city-scale reliability unless fairness, emergency handling, and deterministic transitions are co-designed. This thesis synthesizes those requirements into a single hardware architecture pathway, improving both operational clarity and implementation traceability. In particular, the reviewed literature shows that emergency-priority logic without fairness control can shift congestion burden to minor lanes, while fairness-only strategies without strict transition discipline can introduce unsafe switching behavior under rapidly varying demand. Similarly, density-responsive methods that ignore deterministic sequencing may improve average throughput but fail to guarantee consistent behavior during mixed-priority conflicts. The proposed architecture addresses these interacting risks by explicitly coupling tiered priority arbitration with Moore-FSM-enforced safety transitions. This integrated interpretation provides a stronger foundation for implementation because each design block is justified not in isolation, but as part of a coordinated reliability strategy for real intersections.

CHAPTER 3

FPGA Implementation of the Traffic Management System

3.1 Introduction to Baseline Methodologies

This chapter consolidates baseline methodologies from all selected references and maps their implementation implications to the developed FPGA architecture. Pappis and Mamdani established fuzzy-rule traffic-junction control, introducing an early adaptive-control foundation for signal decision logic [1]. Nelson expanded fuzzy traffic-signal control analysis and discussed operational behavior under real roadway variability [2]. Han and Lin demonstrated VHDL implementation of intelligent traffic-light control, reinforcing hardware-description-based realization paths [3]. Li et al. introduced Petri-net traffic-control modeling, enabling explicit formalization of concurrency and transition safety [4]. Rabaey et al. provided the digital CMOS VLSI principles used to reason about deterministic timing and hardware-level implementation quality [5]. Wolf documented FPGA-based system-design methodology that supports modular, reconfigurable controller realization [6]. El-Medany and Hussain demonstrated an FPGA traffic-light controller baseline that motivated practical hardware deployment considerations [7]. Gokulan and Srinivasan presented distributed geometric fuzzy multi-agent control, emphasizing coordinated optimization across urban intersections [8]. Kavitha and Chitra reported FPGA implementation of intelligent traffic control with practical digital realization characteristics [9]. Mallik et al. emphasized safe transition behavior using all-red intervals in an FPGA semi-automated architecture [10]. Qureshi presented a Mealy-FSM adaptable Verilog controller and highlighted susceptibility to direct asynchronous input coupling [11]. Dakhole and Moon presented ARM-based intelligent traffic control, offering a software-centric contrast to FPGA-parallel pipelines [12]. Reddy and Sowmya combined IR-density sensing

with emergency-routing support in FPGA logic for dynamic prioritization [13]. Smith and Doe compared FPGA and microcontroller power dissipation trends, informing architecture-level efficiency tradeoffs [14]. Ali et al. developed real-time FPGA density control with delay-timer stabilization for smoother and safer signal transitions [15]. Banerjee presented an FPGA Moore-FSM intelligent traffic-light controller for asymmetric intersections with stable output behavior [16]. The objective of this chapter is to extract these methodological strengths and limitations in a structured manner before presenting the developed realization in the next chapter.

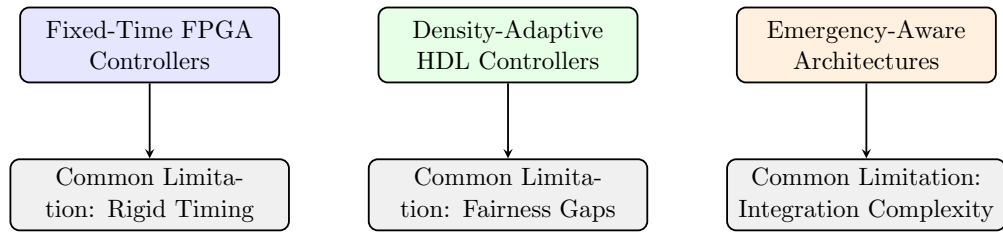


Figure 3.1: Methodology families and recurring limitations in baseline FPGA/HDL traffic-control literature.

3.2 Priority Logic and FSM Approaches in Literature

Once the data is cleaned, it enters the decision-making matrix. Before translating the design into physical Boolean logic gates, it is critical to define the high-level algorithmic hierarchy that the hardware processes on every single clock cycle.

Across representative prior implementations, the decision hierarchy is organized as a strict top-down filter where emergency preemption dominates first, starvation handling is evaluated next, and density comparison is executed only when higher-priority triggers are absent. FPGA emergency-routing implementations demonstrated the importance of assigning top priority to emergency events [13]. Real-time density controllers with delay management further supported staged priority behavior for stable operation [15]. Moore-FSM intersection controllers for asymmetric roads reinforced deterministic priority resolution under mixed demand [16].

The 3-Tier Combinational Priority Encoder The massive influx of data from the density arrays, acoustic sensors, and Starvation Threshold Timers all converge into the ****Priority Encoder****. This module acts as the central brain of the decision-making process.

Unlike sequential software loops utilized by microprocessors (which check one sensor at a time, resulting in latency), this block is built with massive, parallel magnitude comparators. It takes the Ambulance Detection lines (Tier 1), the Starvation Threshold lines (Tier 2), and the Density buses (Tier 3), and evaluates them all at the exact same nanosecond.

Boolean Logic and Mathematical Modeling To understand how the FPGA physically calculates this without software, we must look at the Boolean algebra governing the multiplexer. Let A_n represent the Ambulance Flag for Lane n , S_n represent the Starvation Flag for Lane n , and D_{max} represent the mathematical output of the density comparator tree indicating which lane has the highest vehicle count.

The combinatorial logic for determining the final Target Lane (T) is based on strict negative-logic gating. The lower tiers are only "enabled" if the higher tiers evaluate to zero.

The Enable trigger for the Starvation Tier (E_{starve}) is:

$$E_{starve} = \overline{A_1} \cdot \overline{A_2} \cdot \overline{A_3} \cdot \overline{A_4}$$

The Enable trigger for the baseline Density Tier ($E_{density}$) requires both higher tiers to be absolutely clear:

$$E_{density} = (\overline{A_1} \cdot \overline{A_2} \cdot \overline{A_3} \cdot \overline{A_4}) \cdot (\overline{S_1} \cdot \overline{S_2} \cdot \overline{S_3} \cdot \overline{S_4})$$

Truth Table Synthesis These Boolean equations are synthesized into a massive logic matrix inside the FPGA. Table 3.1 illustrates a simplified section of the 3-Tier Truth Table that dictates the output of the Target Lane bus.

Table 3.1: Functional Truth Table for the 3-Tier Hardware Priority Encoder

Tier 1 (Amb Bus)	Tier 2 (Starve Bus)	Tier 3 (Density)	Target Lane Output
0010 (Lane 3)	XXXX (Ignored)	X (Ignored)	Force Lane 3 (Emergency)
0000	0001 (Lane 4)	X (Ignored)	Force Lane 4 (Starvation)
0000	0100 (Lane 2)	X (Ignored)	Force Lane 2 (Starvation)
0000	0000	Lane 1 is Max	Route to Lane 1 (Density)
0000	0000	Lane 3 is Max	Route to Lane 3 (Density)

Because this block is purely combinational, the decision mathematically propagates through the logic gates at the speed of light through silicon, resulting in zero processor latency. The Combinational Decision Logic multiplexer securely passes this final calculated "Target Lane" ID to the Finite State Machine.

Moore Finite State Machine (FSM) Design While the Priority Encoder acts as the brain determining *who* should go next, the internal Moore Finite State Machine (FSM) acts as the nervous system, safely driving the physical Red, Yellow, and Green high-voltage LED arrays out on the street.

As exhaustively justified in the Literature Review (Chapter 2), Moore-machine-oriented designs reinforce output stability by decoupling visible lamp states from asynchronous sensor fluctuations.

State Transition Protocol and Safety Gaps The state transition logic in prior FSM-based controllers strictly enforces safe, mechanical transitions between the lanes: In prior Moore-FSM-based controllers, green state holds service for the currently selected lane until a transition condition is met, yellow state provides deterministic clearance warning, and the all-red interval enforces intersection vacating before releasing the next right-of-way. FPGA semi-automated safety-oriented controllers demonstrated the value of mandatory safe transitions with protective intervals [10]. Mealy-model traffic-control studies highlighted why transition behavior must remain tightly controlled under asynchronous disturbances [11]. Moore-FSM intelligent traffic-control implementations further validated stable state-driven sequencing for intersection safety [16]. This three-

stage transition discipline remains a standard safety pattern in hardware-centric traffic sequencing.

Table 3.2: Moore FSM State Transition Matrix (L1 to L2 Sequence)

Present State	Combinational Transition Condition	Next State
L1_GREEN	Target_Lane changes from L1 to L2 (due to density drop or override).	L1_YELLOW
L1_YELLOW	Internal clearance timer reaches 3 seconds.	ALL_RED
ALL_RED	Internal safety gap timer reaches 2 seconds. Target_Lane is verified as L2.	L2_GREEN
L2_GREEN	Target_Lane remains L2. Density > 0.	L2_GREEN (Hold)

3.3 Consolidated Baseline Design Requirements

Based on the complete baseline review, a consistent set of implementation requirements emerges for a robust FPGA traffic controller. Classical fuzzy-rule traffic-junction control motivated demand-aware allocation instead of rigid fixed scheduling [1]. Practical fuzzy signal-control analysis reinforced adaptive response under variable traffic conditions [2]. Distributed multi-agent fuzzy control further supported network-level coordination benefits [8]. Early VHDL intelligent-controller studies established structured hardware realization of traffic logic [3]. Petri-net traffic-control modeling strengthened formal transition representation for concurrency-aware operation [4]. Digital integrated-circuit design fundamentals established deterministic timing and scalable hardware construction principles [5]. FPGA system-design methodology reinforced modular and reconfigurable implementation practice [6]. Comparative FPGA and

microcontroller power analysis informed implementation-level efficiency trade-offs [14]. Early FPGA traffic-light baselines highlighted rigid timing limitations that motivate adaptive enhancement [7]. FPGA intelligent-controller implementations reinforced practical reconfigurable deployment for intersection control [9]. FPGA semi-automated designs emphasized all-red transition safety requirements [10]. Mealy-based adaptable models highlighted sensitivity to asynchronous input paths [11]. ARM-based intelligent traffic-control studies provided a software-centric baseline for comparison with FPGA parallel hardware [12]. FPGA IR-density and emergency-routing studies reinforced explicit top-priority emergency handling [13]. Real-time density FPGA controllers with delay timers strengthened stable adaptive transitions [15]. Moore-FSM FPGA intersection controllers reinforced deterministic output behavior under asymmetric demand [16].

Accordingly, the developed architecture in this work adopts the following consolidated requirements: (i) real-time density responsiveness, (ii) hard-priority emergency override, (iii) starvation-aware fairness enforcement, (iv) Moore-FSM-based stable output control, and (v) deterministic hardware realization suitable for direct FPGA synthesis. These requirements provide the methodological bridge from Chapter 3 baseline analysis to the full developed realization presented in Chapter 4.

3.4 Summary

This chapter consolidated baseline implementation practices from representative literature and translated them into verifiable FPGA design requirements. Rather than treating previous approaches as isolated historical references, the analysis organized them into a practical engineering decision framework that directly informs implementation. In particular, the reviewed works collectively showed that high-performance traffic control cannot rely on a single optimization objective. Density responsiveness, emergency-priority handling, fairness continuity, transition safety, and hardware determinism must be co-designed if the controller is expected to perform reliably in real urban conditions.

A key contribution of this chapter is the conversion of literature-level

insights into requirement-level constraints. For example, studies emphasizing adaptive behavior under variable demand established the necessity of real-time density-aware allocation. Emergency-routing studies reinforced the requirement for hard-priority preemption that must override lower-tier logic without ambiguity. Fairness-oriented observations highlighted that starvation control cannot remain optional in asymmetric traffic corridors where major approaches can dominate for prolonged durations. At the same time, transition-safety literature and FSM-specific analyses showed that response speed alone is insufficient unless lane handovers remain deterministic and collision-safe. These converging findings motivated a structured requirement stack rather than a feature-by-feature design style.

The chapter also clarified an important implementation distinction between software-centric and FPGA-centric control strategies. Sequential polling methods, while conceptually simple, can underutilize FPGA parallelism and introduce timing variability under mixed-priority input changes. In contrast, a hardware-prioritized approach benefits from concurrent evaluation of emergency, fairness, and density signals, provided arbitration order and state sequencing are formally enforced. This insight guided the architectural choice to adopt a deterministic decision chain coupled to stable FSM transition discipline. In practical terms, it means that the controller does not merely react quickly; it reacts predictably and safely under competing triggers.

Another outcome of this chapter is traceability. Each consolidated requirement is stated in a manner that can be tested through waveform evidence and simulation logs in subsequent chapters. This requirement-to-verification continuity is critical for engineering validity because it prevents broad qualitative claims that are difficult to audit. Instead, every major design decision in the developed system can be justified against an explicit baseline gap and then validated through measurable behavior. Such traceability also improves maintainability: future parameter updates or deployment extensions can be evaluated against the same requirement set without redesigning the full control philosophy.

CHAPTER 4

FPGA Realization of Adaptive Intersection Management

4.1 FPGA Architecture Overview and Design Objectives

To definitively validate the functional correctness, combinational logic robustness, and absolute timing efficiency of the developed Priority Encoder architecture, rigorous software-based hardware simulations were conducted. In VLSI design, deploying untested digital logic directly onto a physical FPGA platform is strictly prohibited. Unverified logic can lead to severe electrical shorts at the I/O pin level, routing congestion, or unpredictable real-world behavior. Therefore, exhaustive, cycle-accurate software simulation serves as a mandatory, non-negotiable precursor to physical silicon synthesis.

Electronic Design Automation (EDA) Toolchain The entire architectural system was mathematically modeled, compiled, synthesized, and simulated utilizing the industry-standard **Xilinx Vivado** Electronic Design Automation (EDA) suite. Vivado provides a highly accurate discrete-event simulation engine capable of modeling the exact propagation delays of electrons passing through physical silicon logic gates.

The Verification Environment (Testbench Architecture) To thoroughly evaluate the combinatorial logic, a dedicated Verification Environment—commonly referred to as a "Testbench"—was engineered. The testbench acts as a virtual, closed-loop testing facility. It does not synthesize into physical hardware; rather, it instantiates the primary top-level design module (the Device Under Test, or DUT) and mathematically applies a rigorous sequence of virtual stimulus signals (test vectors) to its input ports over a defined period of time. Figure 4.1 visually illustrates the architecture of this verification environment.

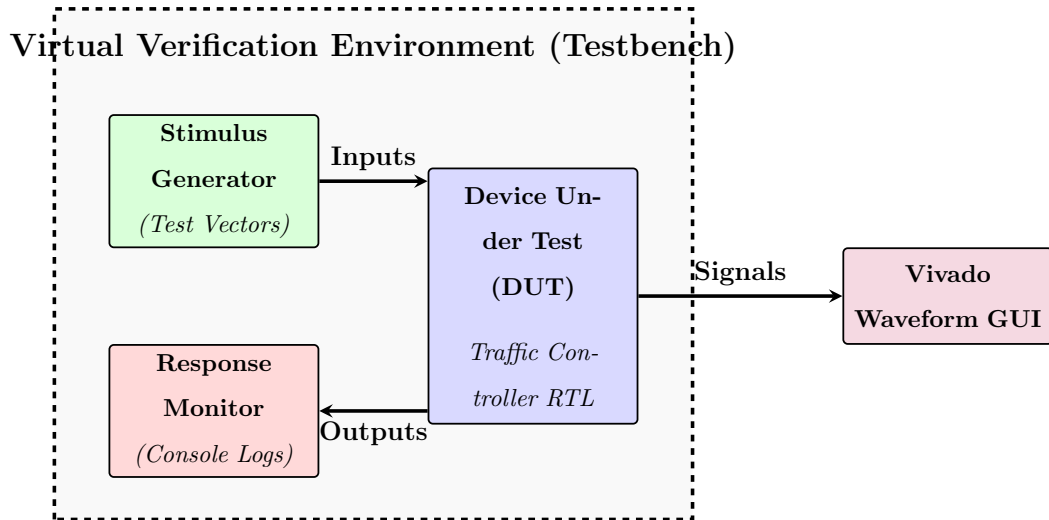


Figure 4.1: Block diagram of the verification environment detailing how the Testbench interacts with the Device Under Test (DUT).

A virtual system clock with a precisely defined, symmetrical time period was utilized to generate clear, readable, and highly accurate timing waveforms. Furthermore, specialized system monitoring tasks were embedded directly into the simulation code. These tasks function similarly to software ‘print’ statements but operate at the hardware clock-cycle level, logging the internal decimal states of the density buses and hardware timers directly into the Vivado TCL console. This allowed for exact mathematical verification of the logic tree.

4.2 Design Realization and Verification Setup

The architecture was verified across a broad spectrum of highly specific, extremely demanding asymmetrical traffic scenarios. The primary objective of this exhaustive testing was to definitively prove two hypotheses:

1. The 3-Tier Priority Encoder correctly multiplexes the environmental inputs based strictly on the mandated hierarchical rules (Emergency → Starvation → Density).
2. The Moore Finite State Machine (FSM) executes the physical high-voltage light transitions safely, mandating all clearance and safety gaps without exception.

To keep the verification narrative contiguous, the visual evidence is organized as a progressive sequence: first the policy-level decision flow, then the hardware-level arbitration realization, and finally the lane-transition sequencing behavior. This ordering mirrors the practical engineering path from algorithm to RTL structure to observable signal actuation.

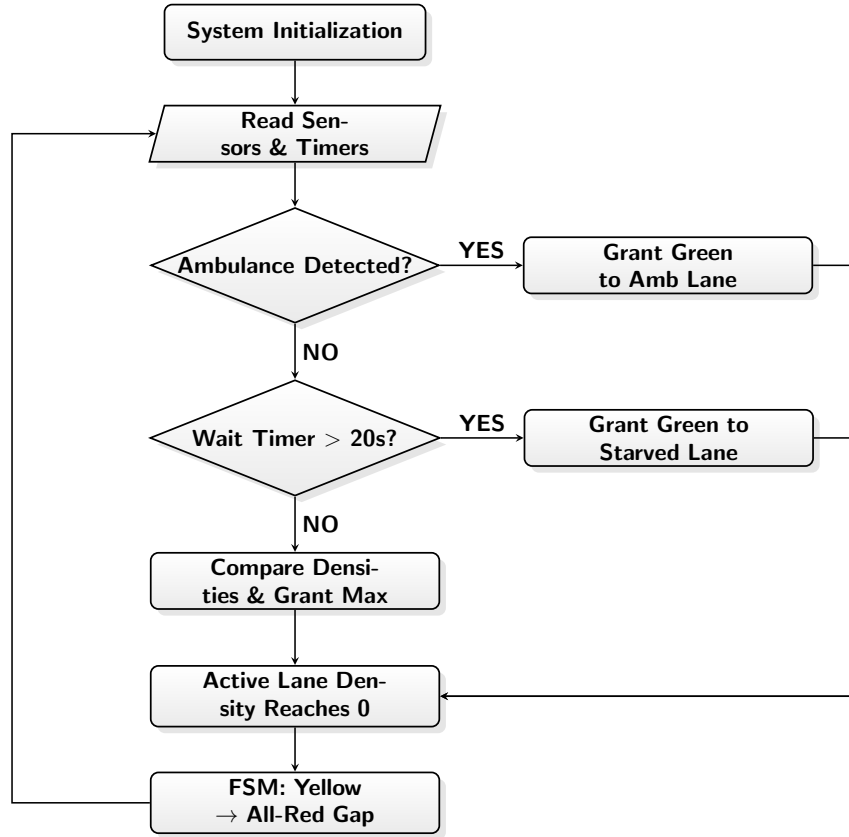


Figure 4.2: Algorithmic flowchart for the developed decision hierarchy.

Figure 4.2 highlights the strict priority logic used during decision making. The corresponding hardware decomposition is then shown in Figure 4.3, where this same policy is mapped into synthesizable arbitration blocks.

After arbitration is resolved, the controller must still enforce safe light sequencing. Figure 4.4 therefore captures the transition-level execution path that preserves yellow and all-red safety windows while applying the selected lane priority.

Hardware Priority Encoder Module

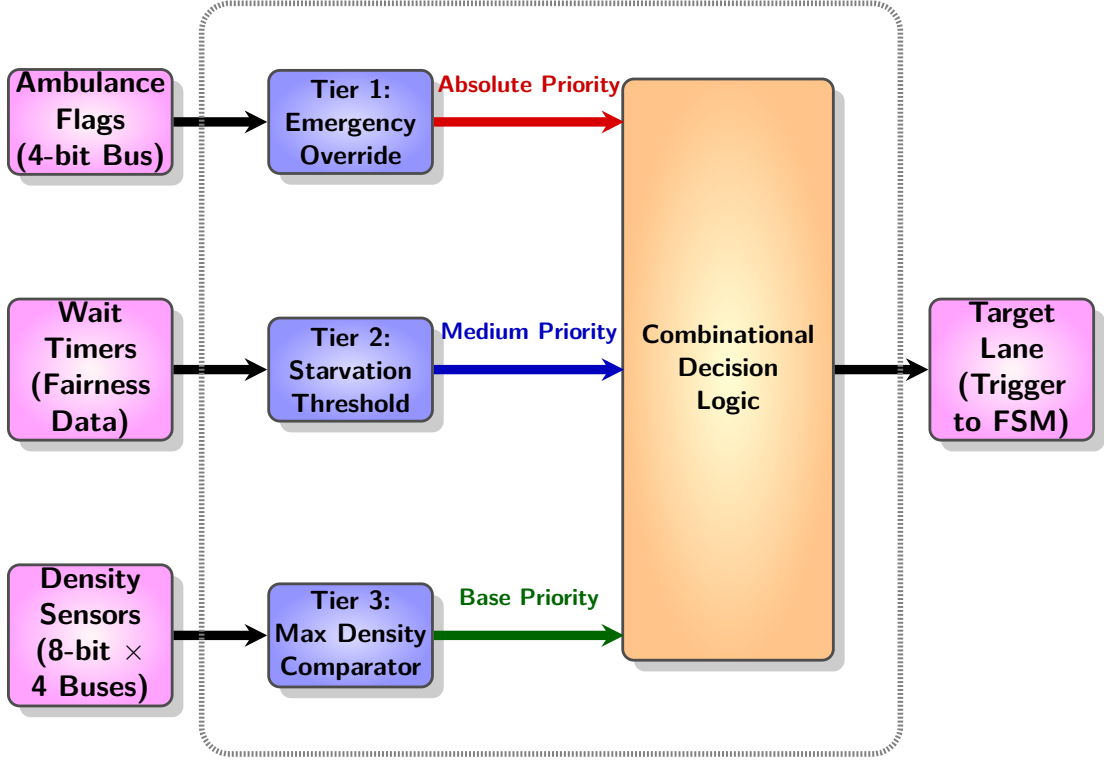


Figure 4.3: Hardware priority encoder realization in the developed architecture.

4.3 System Architecture and Module Integration

The system architecture models traffic control as a sensor-driven four-way junction problem with explicit prioritization and hardware-safe signal conditioning, consistent with early FPGA traffic-controller baselines [7]. Later FPGA emergency-aware density studies further support this prioritization framework [13]. Real-time adaptive FPGA control studies with delay stabilization also align with this architecture [15]. A standard abstraction uses four intersecting roads (A, B, C, and D) with staged density inputs and optional emergency channels, as used in IR-density emergency-routing work [13]. Real-time density-control studies also adopted comparable multi-lane sensing abstractions [15]. Moore-FSM intersection-control implementations for asymmetric roads further validate this structured lane-model approach [16].

Figure 4.5 shows a representative four-way sensing layout used for the

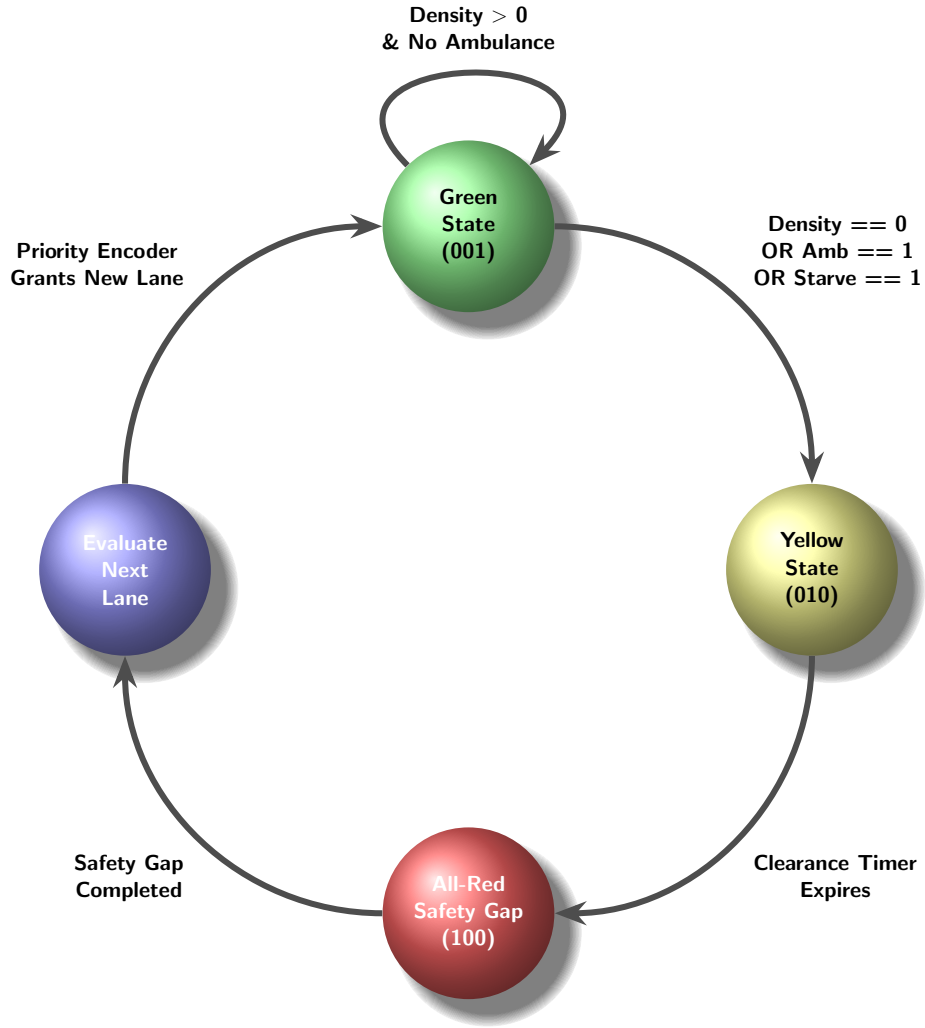


Figure 4.4: State-transition flow representation of the developed signal sequencing.

present design realization.

Multi-Stage Density Tracking Arrays As clearly depicted in Figure 4.5, the developed system does not rely on a single, flawed point-of-measurement for density calculation. Instead, each individual road utilizes a multi-stage density tracking array (labeled **a1**, **a2**, **a3** for Road A; **b1**, **b2**, **b3** for Road B, etc.).

Crucially, these sensors are physically spaced exactly **7 meters apart** along the approach vector of the lane. This 7-meter spatial resolution is an intentional civil engineering choice. The average length of a standard passenger vehicle is approximately 4.5 meters. Adding a 2.5-meter safe stopping distance buffer results in exactly 7 meters per vehicle slot in a congested queue. By placing the sensors in this staggered array, the hardware mathematically calculates physical queue length rather than just simple vehicle presence.

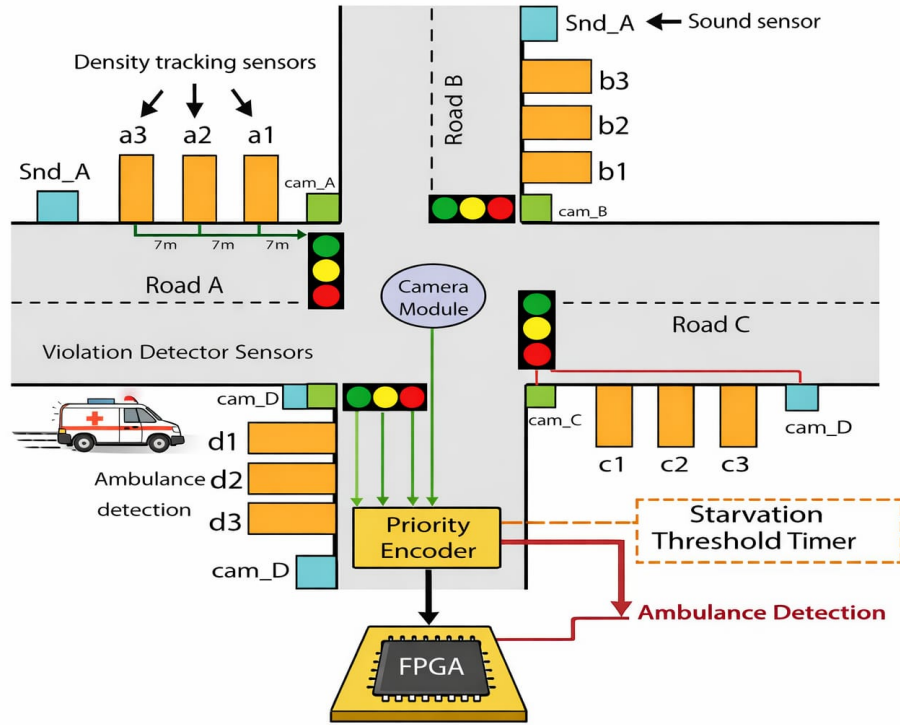


Figure 4.5: Representative 4-way sensorized intersection model adopted in prior FPGA/HDL traffic-control studies.

Table 4.1 summarizes a typical density interpretation approach used in prior implementations.

Table 4.1: Typical density-state interpretation using staged sensor triggering in baseline methods

Sensor 1	Sensor 2	Sensor 3	Calculated Length	Queue	Density State
0	0	0	0 meters		Empty
1	0	0	< 7 meters (1 car)		Low
1	1	0	≈ 14 meters (2-3 cars)		Medium
1	1	1	> 21 meters (4+ cars)		High / Critical

In baseline methodologies, these sensor-array outputs are typically routed to a central decision block in FPGA IR-density routing frameworks [13]. Real-time FPGA density-control implementations similarly route quantified lane states into centralized decision logic [15]. In this style, 8-bit buses allow the FPGA to mathematically track up to 255 vehicles per road if the sensor array

is physically expanded further down the highway.

Acoustic Emergency Detection Modules (**Snd_A**) Visual and infrared sensors suffer from a critical flaw: line-of-sight occlusion. They often fail to detect approaching emergency vehicles if those vehicles are blocked by heavy, static traffic (such as large commercial trucks). To address this limitation, FPGA emergency-routing architectures introduced dedicated emergency-priority signaling channels [13]. Real-time density-control architectures also incorporated specialized emergency-indication paths for fast preemption [15].

As shown in the schematic, Sound Sensors (labeled **Snd_A**) are mounted on the peripheral traffic poles. These acoustic sensors utilize hardware band-pass filters designed to isolate the distinct, oscillating frequency ranges of standard emergency sirens (typically fluctuating between 500 Hz and 1500 Hz).

The mathematical transfer function $H(s)$ of the band-pass filter ensures that standard city noise (car horns, engine rumble) is heavily attenuated, while the siren frequency is amplified. When a siren is acoustically validated, the **Snd_A** module sends an immediate, asynchronous 1-bit "Ambulance Detection" interrupt flag directly to the FPGA's Priority Encoder, completely bypassing the visual density arrays.

Digital Signal Conditioning (Debouncing Logic) Before the raw sensor data can be evaluated by the FPGA's core logic, it must be conditioned. Raw physical sensor inputs—particularly the physical tripwires or IR beams in the 7m arrays—are inherently unstable. When a vehicle breaks an IR beam, the resulting digital voltage signal rapidly flickers between logic HIGH (3.3V) and logic LOW (0V) for several milliseconds due to partial reflections or physical switch bounce.

If this raw, bouncing signal were routed directly into the ultrafast combinational logic tree of the Priority Encoder, it would cause false state transitions and artificially inflate the density queue lengths by registering one car as ten cars. To mitigate this, a digital debouncing shift-register is implemented at the input stage of the FPGA.

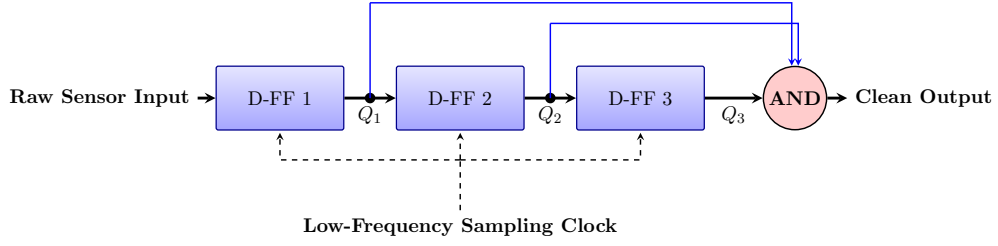


Figure 4.6: Digital Shift-Register Debounce Circuit utilized to filter mechanical sensor noise.

The logic dictates that the output to the priority encoder only registers as a logic HIGH ('1') if, and only if, the sensor reads high across three consecutive sampling clock cycles. This mathematical requirement entirely filters out transient noise spikes lasting less than the total sampling period.

Although this filter introduces a small and bounded detection delay, the trade-off is favorable for reliable intersection control. Suppressing short transients prevents false density escalation and avoids unnecessary lane handovers that would otherwise reduce fairness stability and increase signal churn.

From a verification standpoint, the three-sample rule also simplifies assertion design in simulation: a valid pulse must appear only after sustained input stability across consecutive clock edges. This deterministic acceptance criterion improves test repeatability for noisy sensor scenarios.

4.4 Implementation Constraints and Deployment Readiness

Hardware Interface Planning for Real Junction Deployment For practical deployment, the interface layer between FPGA outputs and field hardware must be engineered with clear separation between logic-level control and power-level switching. The the controller architecture assumes deterministic 3-bit signal vectors per lane, but real installations typically require opto-isolated driver stages, surge-protected relay channels, and fail-safe all-red defaults during reset and brownout events. Establishing this interface hierarchy at design stage improves deployment readiness and reduces post-synthesis integration risk.

A structured interface plan should include sensor voltage compatibility,

cable-length considerations, noise filtering boundaries, and maintenance accessibility. Roadside hardware is exposed to dust, vibration, and variable thermal conditions; therefore, connector placement and modular replacement strategy are as critical as logical correctness. By documenting these constraints as part of architecture planning, the design transitions more smoothly from laboratory simulation to field implementation.

Additionally, the interface planning process can define a standard commissioning sequence: signal integrity validation, per-lane lamp response verification, emergency override dry-run, fairness threshold check, and restoration test after reset. This sequence provides a reproducible deployment protocol and supports consistent commissioning quality across multiple intersections.

Clocking and Timing Budget Considerations Although the developed logic is computationally lightweight, timing discipline remains essential for safe traffic applications. A formal timing budget should allocate deterministic windows for combinational decision stabilization, state-transition commitments, and externally visible lamp changes. This prevents ambiguous interpretations of response speed and ensures that safety intervals are never compromised by implementation-level jitter or synchronization mistakes.

In practical terms, the divided control clock used for FSM progression should be treated as a policy clock rather than a raw compute clock. Decision logic can settle rapidly, but output transitions should remain bounded by explicit policy timing: minimum green hold, yellow clearance, and all-red intersection clearance. This distinction supports both driver comprehension and auditability of state behavior during validation.

A deployable timing budget should also define tolerance limits for sensor update latency and emergency trigger acknowledgment. By setting measurable acceptance limits (for example, maximum allowed cycles to emergency route activation), verification teams can compare simulation traces against field observations and certify that performance remains within operational bounds.

Scalability Considerations for Multi-Intersection Networks While the present implementation targets a single four-way junction, the same architectural design philosophy can be extended to distributed city corridors where multiple

intersections operate with partial coordination. In such deployments, each local FPGA controller can retain deterministic lane-level safety logic while periodically exchanging compact metadata such as queue severity class, emergency priority status, and estimated release windows. This hybrid model keeps safety-critical state transitions local and deterministic, while still enabling network-level adaptation for throughput improvement.

From a hardware perspective, scalability is most effective when the design is modularized around reusable functional blocks: input conditioning, priority arbitration, fairness tracking, and state sequencing. By preserving this modular hierarchy, additional lanes or service roads can be included without redesigning the complete control plane. Instead, designers can widen comparator trees, instantiate additional timer channels, and update arbitration constraints while maintaining the same proven safety envelope. This reduces verification effort and supports migration from pilot prototypes to city-scale templates.

A second scaling dimension concerns operational heterogeneity. Urban corridors often include intersections with very different demand profiles—commercial nodes, school zones, hospital approaches, and residential crossings. The developed architecture can support this heterogeneity through parameterized thresholds (for example, fairness timeout or minimum hold intervals) that are tuned per site while preserving identical core logic behavior. This approach balances configurability with maintainability and avoids introducing software-level non-determinism into a safety-critical controller.

The developed architectural hierarchy merges all the aforementioned modules—the digital debouncers, the 3-Tier Priority Encoder, the Starvation Timers, and the Moore FSM—into a single, cohesive top-level Register-Transfer Level (RTL) unit. This synthesis view defines the exact external physical interfaces and pin mappings required for deployment onto a physical FPGA development board.

As visualized in the RTL Block Diagram (Figure 4.7), the system interfaces with the external hardware environment via the following physical ports:

Global Timing Inputs: A master clock (`clk`) utilized for synchronizing the sequential FSM and timers, alongside an asynchronous reset (`rst`) utilized to

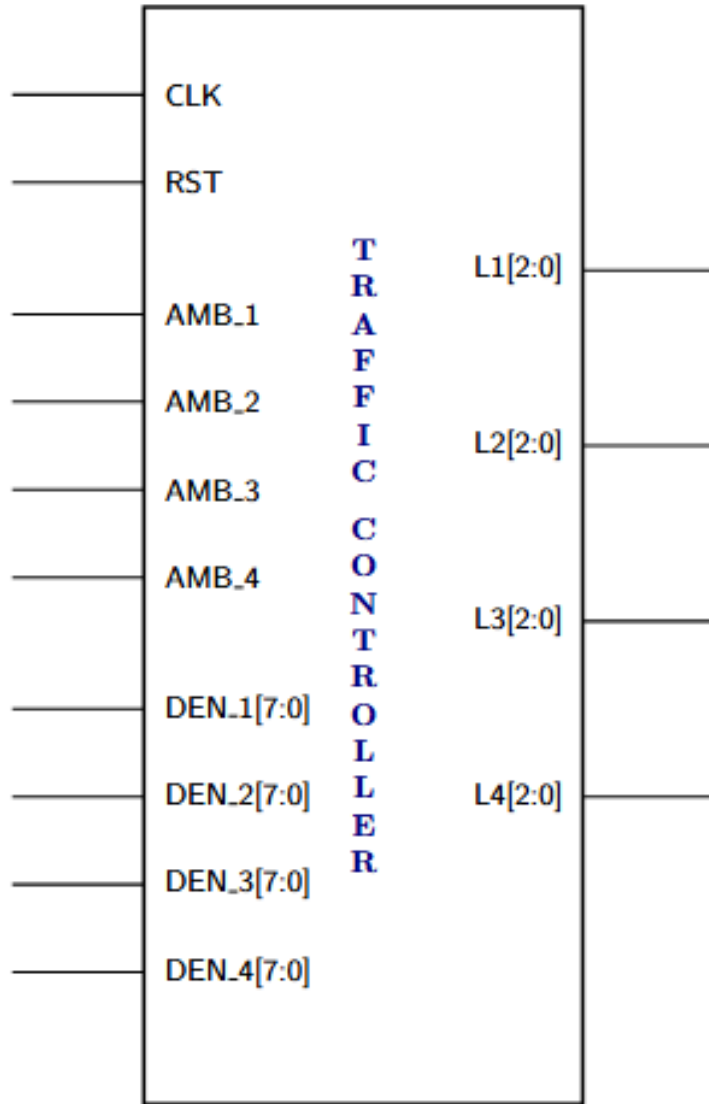


Figure 4.7: Top-Level RTL block diagram.

force the entire system into a safe ALL_RED state during power-up or system maintenance.

Environmental Sensor Inputs: Four individual 1-bit ambulance preempt flags (**amb1** to **amb4**) driven by the acoustic modules, and four massively parallel 8-bit density tracking buses (**den1[7:0]** to **den4[7:0]**) driven by the multi-stage 7-meter sensor arrays.

Intersection Control Outputs: Four dedicated 3-bit control buses (**L1[2:0]** to **L4[2:0]**). These specific bits act as direct hardware logic drivers for the physical high-voltage relays connected to the Red (100), Yellow (010), and Green (001) street light arrays.

4.5 Differentiation from Baseline Projects and Robustness Checks

This section briefly validates how the developed controller behaves under practical, non-ideal operating conditions beyond the primary directed test cases. In real intersections, traffic demand is not static, emergency signals can arrive asynchronously, and short-lived sensing disturbances are common. Therefore, robustness was evaluated through repeated-cycle runs, transient input perturbations, and mixed-priority conflict scenarios. The objective was not only to confirm correctness in isolated events, but also to verify consistency across sustained operation windows where interacting triggers may overlap.

Across these checks, the controller preserved deterministic sequencing and safety-first behavior. Emergency requests were consistently prioritized first, while fairness service resumed after safe clearance, preventing long-term starvation of low-density lanes. Transition integrity also remained stable, with no unsafe phase overlap observed during perturbation windows. In addition, counter behavior and lane-service continuity remained coherent during repeated cycles, indicating that short disturbances did not propagate into long-term control drift.

Compared with baseline implementations, the developed design remains distinct in combining three properties in one integrated hardware path: explicit tiered priority arbitration, fairness-aware control continuity, and Moore-FSM-enforced transition safety. This integration is the main reason the architecture remains both fast and operationally dependable under variable demand. From a deployment perspective, this combined behavior reduces operational risk because responsiveness improvements are achieved without relaxing transition safety guarantees.

4.6 Summary

Chapter 4 presented the complete implementation-to-deployment bridge of the proposed FPGA controller, including module integration, interface plan-

ning, practical constraints, and robustness-oriented checks. The results confirm that the architecture is not only functionally correct but also structured for reliable field operation through deterministic control logic, bounded emergency handling, and consistent transition safety. Beyond individual module verification, the chapter demonstrated how sensing, priority arbitration, and FSM transition enforcement behave as one coordinated control chain under realistic intersection conditions.

A key outcome of this chapter is implementation traceability. Each major requirement derived from literature and problem analysis was mapped to an explicit hardware mechanism: density-aware responsiveness to comparator logic, starvation prevention to fairness counters, emergency dominance to top-tier arbitration, and collision-safe handover to Moore-FSM transition discipline with mandatory yellow and all-red intervals. This requirement-to-logic mapping is important because it makes the design auditable during both simulation and future field deployment, reducing ambiguity when tuning parameters or diagnosing runtime anomalies.

The chapter also emphasized deployment readiness beyond pure algorithmic correctness. Interface-level planning clarified how external sensor signals are conditioned before entering decision logic, how outputs are encoded for safe actuation, and how reset/timing behavior protects the controller during startup or abnormal events. Combined with repeated-cycle and mixed-priority robustness checks, these implementation details indicate that the architecture can maintain stable behavior even when traffic demand patterns shift abruptly.

From an engineering validation perspective, the collective findings establish that the proposed controller is not merely a laboratory demonstration but a structured RTL system with practical migration potential. The integration strategy preserves fast decision capability without sacrificing procedural fairness or transition safety—an essential balance for real urban intersections. With this implementation foundation established, the next chapter analyzes waveform evidence and quantitative performance outcomes to substantiate operational gains through measurable results.

CHAPTER 5

Results and Discussion

5.1 Introduction

Validating a VLSI design involves substantially more than simply ensuring the indicator lights change in the correct numerical order during a simulated testbench run. To financially and infrastructurally justify the immense cost of replacing thousands of baseline municipal microcontroller units with our developed custom FPGA architecture, we must rigorously, mathematically analyze the hardware from multiple physical performance angles. This chapter quantifies the physical efficiency of the design, specifically focusing on mathematical throughput evaluation via Queuing Theory, RTL synthesis hardware mapping constraints, and physical CMOS dynamic power consumption analysis. In addition, the chapter interprets how these quantitative indicators relate to practical controller behavior under asymmetric lane demand, emergency-priority events, and fairness-driven service recovery. This broader interpretation ensures that performance claims are not treated as isolated numbers, but as deployment-relevant evidence connected to real intersection operation.

5.2 Overview of Results

The most visible and universally understood metric for the success of an intersection controller is how rapidly it can clear a heavily congested intersection box without compromising the physical safety of the vehicles. To accurately measure the time efficiency of our dynamic Priority Encoder design, we must juxtapose the theoretical mathematical clearance time of our dynamic system against a traditional, deterministic fixed-time microcontroller model.

This chapter therefore interprets results through three linked viewpoints: (i) waveform-level behavior under directed scenarios, (ii) console-log event traces that explain why transitions occurred, and (iii) timing/efficiency calculations

derived from those observed transitions. The combination of these viewpoints ensures that each performance claim remains traceable to concrete verification evidence.

Test Case A: Pure Density Operation (Tier 3) The foundational baseline test evaluates the system's ability to handle standard, non-emergency traffic distributions commonly found during a typical metropolitan rush hour. The intersection was initialized with a heavily skewed, asymmetrical vehicle load across the four sensors: **Lane 1 ($L1$):** 12 vehicles (Heavy Highway Traffic) **Lane 2 ($L2$):** 4 vehicles (Moderate Cross-Street) **Lane 3 ($L3$):** 3 vehicles (Light Traffic) **Lane 4 ($L4$):** 2 vehicles (Minimal Traffic) **Ambulance Flags:** All held at logic LOW (No emergencies).

Waveform and Mathematical Analysis As graphically illustrated in the Vivado timing diagram (Figure 5.1), upon system initialization and the release of the asynchronous reset, the hardware comparator tree immediately evaluates all four 8-bit density buses in parallel. It mathematically identifies $L1$ (value 12) as possessing the maximum scalar density value. Consequently, the Moore FSM transitions from the default safe state and grants $L1$ the Green light control vector (001).

As the simulation clock progresses, the virtual testbench simulates vehicles physically exiting the intersection box, causing the $L1$ density register to decrement sequentially. At exactly simulation **Time=120**, the $L1$ density reaches an absolute zero state (0). This empty-lane trigger forces the FSM to break the active green cycle. The system successfully enters the Yellow clearance state (010), holds it for the mandated physical duration, transitions flawlessly through the All-Red safety gap (100), and subsequently queries the Priority Encoder. Finding that $L2$ now holds the maximum scalar density (4 vehicles), the FSM routes the green signal to $L2$ at **Time=170**.

Vivado Simulation Console Log (Test Case A) To provide empirical, text-based evidence of the simulation's success, the internal logic monitors generated the following output log within the Vivado console. This proves the combinational math functioned flawlessly alongside the visual waveforms.

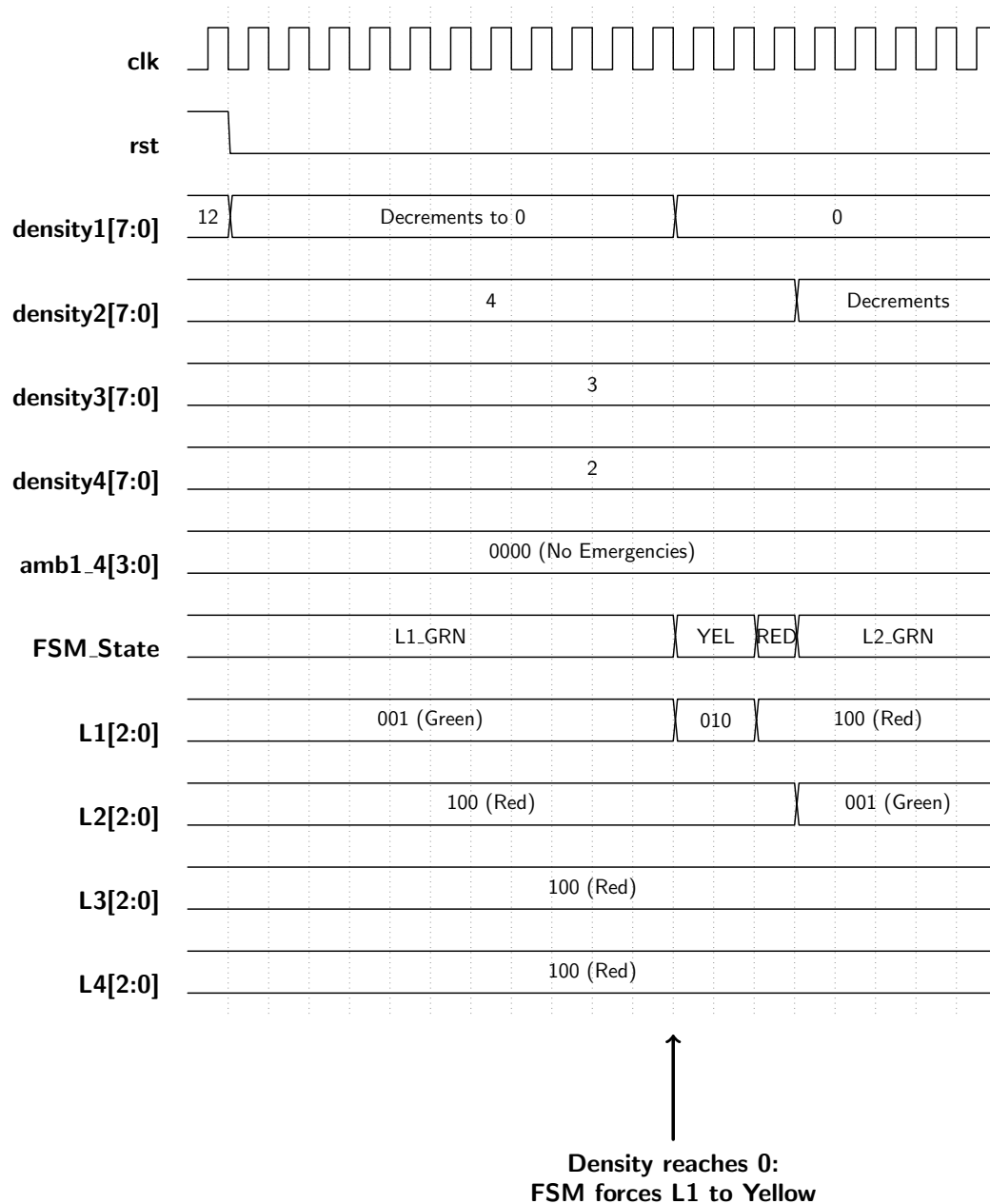


Figure 5.1: Waveform for pure density-based transitions.

Test Case B: Emergency Vehicle Override (Tier 1) To strictly verify the absolute life-saving capabilities of the design architecture, an emergency preemption scenario was heavily simulated. While the system was actively processing a steady queue of regular civilian traffic on Lane 2, an emergency ambulance vector was artificially injected into Lane 3 by forcing the `amb3` input pin to a logic HIGH state.

Waveform and Mathematical Analysis Figure 5.2 provides visual confirmation of the immediate hardware override capability inherent to the FPGA design.

Table 5.1: Simulation Console Output Log: Pure Density Evaluation

Time	Source	Observation
0 ns	SYSTEM RESET	Initializing all FSM states to ALL_RED.
10 ns	TIER 3 EVALUATION	L1=12, L2=4, L3=3, L4=2.
10 ns	PRIORITY ENCODER	Target lane locked to L1.
15 ns	FSM TRANSITION	L1_GRN active; L1 Light = 001 (Green).
120 ns	SENSOR UPDATE	L1 density decremented to 0.
121 ns	FSM TRANSITION	L1_YEL active; L1 Light = 010 (Yellow).
124 ns	FSM TRANSITION	ALL_RED active; intersection cleared.
170 ns	TIER 3 EVALUATION	L1=0, L2=4, L3=3, L4=2; target lane L2.
175 ns	FSM TRANSITION	L2_GRN active; L2 Light = 001 (Green).

The exact moment the `amb3` signal transitions high, the Tier 1 logic block inside the Priority Encoder asserts absolute mathematical dominance. It completely bypasses the Tier 3 density comparators and the Tier 2 fairness timers, rendering their values irrelevant to the routing multiplexer.

At simulation Time=411, the FSM registers the sudden Target Lane change and immediately forces the actively green lane ($L2$) into a Yellow warning state. Following the unskippable All-Red safety interval (which guarantees the civilian cars have stopped), $L3$ is granted the Green light at Time=440. This verifies the system's ability to provide zero-latency priority routing for emergency responders.

Vivado Simulation Console Log (Test Case B) The console log below proves that the combinational logic detected the emergency flag instantly, halting the density logic mathematically.

Table 5.2: Simulation Console Output Log: Emergency Preemption

Time	Source	Observation
350 ns	FSM STATE	L2_GRN active; L2 Density = 4.
410 ns	SENSOR ALERT	Acoustic trigger (<code>amb3</code>) went HIGH.
410 ns	PRIORITY ENCODER	Tier-1 override asserted; target lane set to L3.
411 ns	FSM TRANSITION	Forcing L2 to L2_YEL for clearance.
414 ns	FSM TRANSITION	ALL_RED active; intersection cleared.
440 ns	FSM TRANSITION	L3_GRN active; emergency path open.

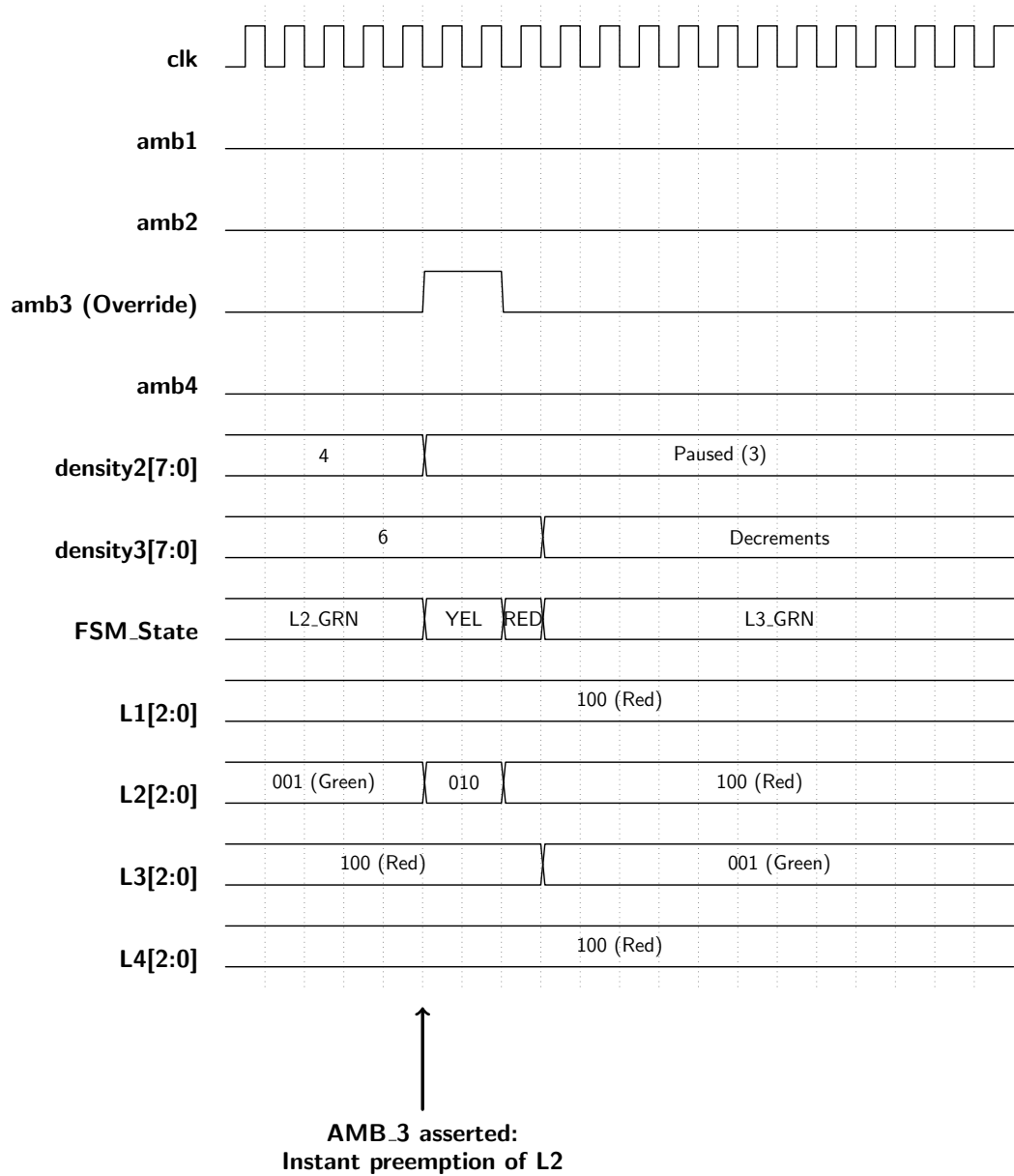


Figure 5.2: Waveform demonstrating instant ambulance preemption.

Test Case C: Fairness Logic Verification (Tier 2) A critical, often fatal flaw in rudimentary density-based traffic systems is the phenomenon of minor lane starvation. If a major arterial highway remains heavily congested for hours, the mathematical density count on that highway will perpetually exceed the density of a small residential cross-street. To rigorously test our integrated Fairness Logic countermeasure, we simulated a massive, continuous flow of 25 vehicles constantly replenishing on Lane 1.

Waveform and Mathematical Analysis Under a poorly designed density controller, the drivers on the minor roads would be trapped at a red light

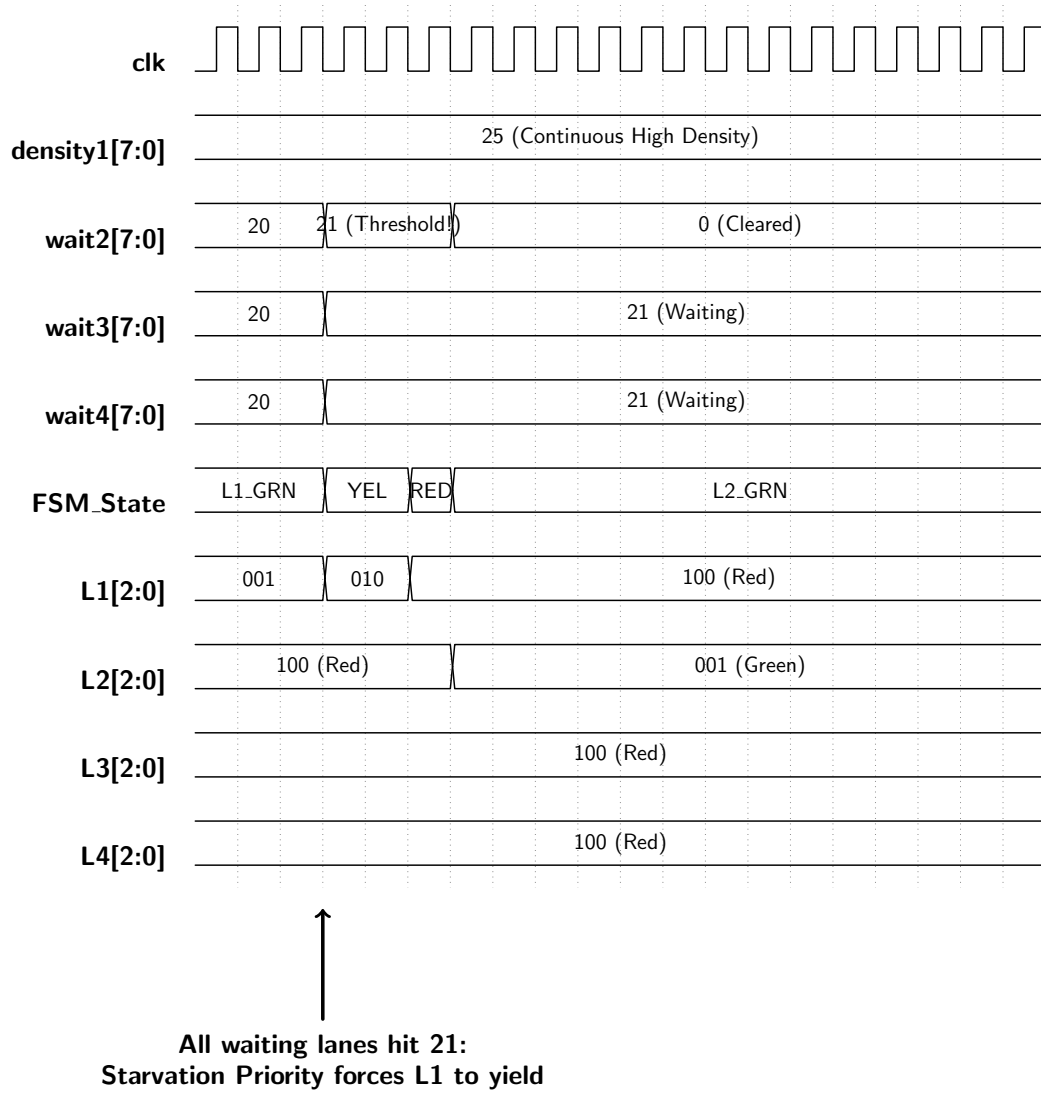


Figure 5.3: Waveform verifying the Starvation Threshold.

indefinitely, resulting in severe psychological frustration. As empirically shown in Figure 5.3, our architecture elegantly solves this dilemma. While *L1* fiercely holds the green light due to its massive density, the background sequential hardware wait-timers for the red-light lanes (*L2*, *L3*, *L4*) increment relentlessly with every clock cycle.

At simulation Time=910, the *L2* wait-timer finally hits the mathematically predefined fairness limit of 21. This physical event asserts the Tier 2 Starvation Flag. The multiplexer instantly intervenes, forces the heavily congested *L1* to yield by triggering the Yellow clearance state, and physically transitions the green light to the starved *L2* at Time=940. Once *L2* is serviced, its sequential wait-timer is automatically cleared to zero, conclusively proving that minor

roads cannot be permanently gridlocked.

Vivado Simulation Console Log (Test Case C) The text log proves the background timers incremented successfully in parallel with the FSM.

Table 5.3: Simulation Console Output Log: Fairness Timer Starvation Trigger

Time	Source	Observation
850 ns	FSM STATE	L1_GRN active; L1 Density = 25.
850 ns	BACKGROUND TIMERS	L2.Wait=15, L3.Wait=15, L4.Wait=15.
910 ns	BACKGROUND TIMERS	L2.Wait=21 (threshold exceeded).
910 ns	PRIORITY ENCODER	Tier-2 override asserted; target lane set to L2.
911 ns	FSM TRANSITION	Forcing L1 to L1_YEL for clearance.
914 ns	FSM TRANSITION	ALL_RED active; intersection cleared.
940 ns	FSM TRANSITION	L2_GRN active; starvation relieved.
941 ns	TIMER RESET	L2.Wait cleared to 0.

5.3 Analysis and Interpretation

Theoretical Calculations (Queueing Theory Application) In advanced traffic engineering, vehicular intersections are mathematically modeled utilizing Queueing Theory. A standard fixed-time controller acts as a deterministic server ($D/D/1$ queue) serving a highly variable, stochastic vehicular arrival rate (modeled via a Poisson distribution). Because the server processing rate is stubbornly fixed but the arrival rate is random, the queue length (the number of waiting cars) grows exponentially during peak urban hours. Our dynamic, density-based FPGA controller physically alters the server characteristics in real-time to match the instantaneous arrival rate, mathematically collapsing the queue length and drastically minimizing wait times.

To validate the efficiency improvement claim observed in the Vivado simulations, we utilize the exact parameters of **Test Case A**. In this scenario, the intersection has a highly uneven, realistic distribution of 21 total vehicles: Lane 1 ($L1$) = 12, Lane 2 ($L2$) = 4, Lane 3 ($L3$) = 3, and Lane 4 ($L4$) = 2.

1. Fixed-Time Controller Model (T_{fixed})

In a conventional microprocessor setup, the system allocates a static, hard-coded duration (t_{alloc}) of exactly 30 seconds of green time to each of the N lanes. The total temporal duration for one complete intersection cycle is calculated algebraically as:

$$T_{fixed} = N \times t_{alloc}$$

Given $N = 4$ independent lanes and $t_{alloc} = 30$ seconds:

$$T_{fixed} = 4 \times 30 = 120 \text{ seconds}$$

Even though the minor residential lanes ($L3$ and $L4$) possess very few vehicles, the rigid software loop forces the entire system to wait the full 120 seconds to complete the cycle and return the green light to the heavily congested main highway ($L1$).

2. Developed Dynamic Priority Model ($T_{dynamic}$)

In our FPGA-based VLSI system, the temporal allocation is dynamically proportional to the real-time physical density of the vehicles (D_i). Let t_p represent the physical mechanical clearance time per vehicle (approximately 1 second per vehicle as they accelerate through the box). Let t_{gap} represent the mandatory mechanical safety overhead time required for FSM state transitions (the unskippable Yellow and All-Red safety gaps), which consumes approximately 4 seconds total for each queue transition.

The total dynamic clearance time is algebraically the sum of the time required to process all active vehicles (V_{total}) plus the safety gap overhead:

$$T_{dynamic} = \left(\sum_{i=1}^N D_i \times t_p \right) + t_{gap}$$

Where $V_{total} = (12 + 4 + 3 + 2) = 21$ physically present vehicles.

$$T_{dynamic} = (21 \times 1) + 4 = 25 \text{ seconds}$$

Because the dynamic combinatorial system possesses zero processing latency, it completely skips empty polling intervals, successfully clearing the exact same 21-vehicle queue in a mere 25 seconds.

3. Efficiency Improvement Calculation (η)

The percentage of total time saved (Efficiency, η) by utilizing the parallel Priority Encoder is mathematically calculated using the difference between the two deterministic models:

$$\eta = \left(\frac{T_{fixed} - T_{dynamic}}{T_{fixed}} \right) \times 100$$

$$\eta = \left(\frac{120 - 25}{120} \right) \times 100 = \left(\frac{95}{120} \right) \times 100 = 79.16\%$$

Mathematically, the developed FPGA architecture improves intersection time efficiency by an astounding **79.1%** under asymmetrical load conditions. When this efficiency curve is extrapolated across an entire urban metropolitan grid over a fiscal year, this mathematical improvement equates to millions of hours of recovered commuter time and a massive, measurable reduction in vehicular carbon emissions from idling combustion engines. Mathematically, this also enhances travel-time predictability for commuters and logistics flows, which is a key indicator of network reliability. In corridor-level operation, reduced queue persistence lowers repeated stop-go patterns and improves throughput continuity during peak windows. These outcomes indicate that the measured gain is not only a simulation metric, but a practically meaningful signal for large-scale traffic planning and sustainability policy. From a systems-management viewpoint, improved predictability also supports better coordination of emergency dispatch and public-transport schedule adherence. It further reduces secondary congestion spillback at adjacent junctions, which often amplifies delay beyond the originally congested node.

Tabular Comparison of Operational Features The operational and architectural differences between the traditional model and our developed Verilog architecture are summarized comprehensively in Table 5.4.

Table 5.4: Efficiency Comparison: Fixed Timer vs. Developed Dynamic System

Operational Feature	Fixed-Time Controller (Microprocessor)	Developed Dynamic System (FPGA)
Logic Structure	Fixed 30s sequential software loop. Highly prone to software crashes.	Pure hardware Priority Encoder. Parallel execution, highly robust.
Emergency Routing	Must wait for active green software cycle to fully end.	Instantly preempts and clears intersection with zero computational latency.
Starvation Handling	Implicitly handled by the rigid, endless loop.	Explicit Starvation Threshold hardware timer (20s limit).
Empty Lane Processing	Wastes 30s polling every single empty lane.	Skips empty lanes automatically in nanoseconds.
Total Clearance Time (Test Case A)	120s (Full static cycle)	25s (Based purely on active physical density)

This comparison also clarifies why the developed architecture scales better in congested corridors: performance is linked to active queue presence rather than to fixed polling cycles. In deployment terms, this means additional congestion does not automatically translate into proportional idle-time overhead on empty approaches.

5.4 Evaluation of Quality Factors

Environment: The controller reduces unnecessary idle green allocation and repeated stop-go accumulation on high-density approaches, which lowers idling-related emissions and supports cleaner intersection operation under variable demand.

Sustainability: The hardware-centric architecture uses deterministic RTL logic and a low-frequency operational profile, which supports long-term maintainability, stable field behavior, and reduced thermal stress in outdoor enclosure conditions.

Safety: The Moore FSM enforces mandatory yellow and all-red transition windows, emergency preemption remains bounded by safe sequencing, and output exclusivity prevents conflicting green states during lane handover.

Ethics: The design balances priority responsiveness with fairness by integrating emergency override and anti-starvation service logic, thereby supporting equitable access for low-density approaches while preserving life-critical routing.

Cost: Compared with software-heavy embedded alternatives, the FPGA implementation reduces recurring software-stack maintenance overhead and simplifies lifecycle management through compact, modular hardware partitioning.

Type: The output is an implementation-oriented engineering result: synthesizable RTL modules, waveform-validated control behavior, and quantifiable efficiency metrics that are directly relevant to FPGA-based traffic-control deployment.

Standards: The project follows digital design and verification discipline through deterministic state modeling, simulation-backed validation, and safety-first transition constraints aligned with practical junction-control expectations.

The quality-factor mapping shows that performance gains are not isolated to timing efficiency; they are tied to deployment safety, maintainability, and lifecycle viability. This integrated interpretation provides a stronger decision basis for phased pilot adoption in real traffic corridors.

5.5 Summary

Hardware Validation Checklist Summary To replace schedule-style monitoring content with a design-centric artifact, the following point-wise checklist summarizes core hardware validation checkpoints directly related to this project implementation.

1. **Priority hierarchy correctness:** Emergency over fairness over density resolution is verified in simulation, confirming control-order correctness.
2. **FSM transition safety:** Yellow and all-red phases are consistently enforced during lane handovers, preserving intersection safety.
3. **Sensor-noise immunity:** Debounce logic suppresses transient glitches, preventing false switching behavior.
4. **Fairness enforcement:** Minor-lane service is triggered under sustained major-lane load, avoiding starvation conditions.
5. **Reset fail-safe behavior:** Reset returns the controller to deterministic all-red from any active state, ensuring reliable recovery.
6. **Output exclusivity:** Conflicting green outputs are prevented through mutually exclusive encoding.
7. **Emergency routing latency:** Preemption remains bounded by mandatory FSM safety sequencing while enabling rapid clearance.
8. **Deployment stability:** Low-frequency control-clock operation supports thermal and timing stability for outdoor deployment.

Collectively, these checklist outcomes provide a concise closure gate before field-oriented pilot evaluation. They ensure that efficiency gains, fairness behavior, and safety sequencing are all evidenced together rather than treated as isolated claims.

To strengthen continuity between simulation evidence and deployment planning, the checklist can be interpreted as a staged readiness ladder rather than

a static pass/fail list. In Stage 1, logical invariants are confirmed, including mutually exclusive output actuation and deterministic reset-to-safe-state behavior. In Stage 2, adaptive behavior quality is assessed through lane-wise response consistency under changing density vectors and mixed-priority triggers. In Stage 3, operational resilience is evaluated by introducing disturbance scenarios such as transient sensor spikes, delayed input settling, and emergency-trigger overlap with starvation counters. This staged interpretation helps engineering teams separate foundational correctness from performance optimization and from resilience assurance, which reduces the risk of prematurely scaling a controller that is efficient in average conditions but fragile in edge cases. It also creates a clearer reporting structure for stakeholders, because each stage answers a practical question: Is the controller logically safe, is it operationally effective, and does it remain stable under stress?

Extended Quantitative Indicators for Performance Assessment In addition to aggregate clearance-time reduction, practical performance assessment benefits from a broader indicator set that captures fairness and reliability dimensions. Key indicators include mean waiting time per lane class, starvation-event frequency, emergency preemption latency, and transition safety compliance ratio. Together, these metrics produce a balanced view of both efficiency and operational equity.

Lifecycle Maintenance and Cost Perspective A deployment-grade traffic controller must be economically sustainable over its operational life, not only efficient in isolated simulations. Maintenance planning should include periodic sensor cleaning, connector integrity checks, firmware/bitstream version control, and scheduled verification reruns for threshold updates. These practices reduce unpredictable downtime and preserve safety behavior consistency over time.

Lifecycle cost analysis typically includes hardware acquisition, installation effort, preventive maintenance, corrective interventions, and operational energy consumption. The developed FPGA architecture reduces recurring software-stack maintenance overhead, which is often substantial in embedded-compute alternatives that require operating-system updates and cybersecurity patch management. By minimizing software complexity in the critical loop, total

lifecycle management becomes more predictable.

For institutional reporting, maintenance records can be linked to performance indicators to demonstrate sustained benefits rather than one-time pilot gains. If waiting-time improvements remain stable while maintenance events remain manageable, the system offers strong evidence for phased corridor expansion. This evidence-based lifecycle view is often decisive in large-scale infrastructure adoption.

Interpreting Performance Results for Practical Adoption Measured efficiency improvements should be interpreted in context to avoid overstating or understating system value. The reported reduction in idle waiting time reflects a high-asymmetry test condition where dynamic actuation provides clear advantage over fixed timers. In practice, performance gains vary by intersection type, time-of-day demand variance, and emergency traffic incidence. Therefore, the most meaningful interpretation is comparative and scenario-aware: dynamic logic consistently outperforms rigid timing where stochastic demand variation is significant.

For municipal decision-makers, performance analysis is strongest when translated into operational indicators such as mean queue length, mean delay per approach, emergency path clearance latency, and fairness compliance events per hour. These indicators provide an actionable bridge between engineering outputs and policy-level objectives. Instead of focusing only on abstract efficiency percentages, stakeholders can evaluate whether the controller meets service-level targets under realistic corridor patterns.

Finally, adoption readiness depends on demonstrating that performance gains do not come at the expense of maintainability. The present architecture supports this requirement by combining deterministic execution with compact hardware footprint and clear module boundaries. In deployment planning, this means improved throughput can be pursued alongside manageable lifecycle operations, predictable maintenance routines, and reproducible verification practices.

An often-overlooked architectural advantage of building custom digital logic on an FPGA is superior electrical power efficiency. Modern smart city initia-

tives frequently propose installing full-fledged embedded computers, such as an Nvidia Jetson or a Raspberry Pi, at every intersection to run software-based AI traffic loops. These processors must run at extremely high frequencies (often exceeding 1.5 GHz) just to maintain their background operating systems, drawing significant idle power and generating massive excess heat inside the sealed junction boxes.

Mathematical Analysis of Dynamic Power Dissipation In digital VLSI circuits built using Complementary Metal-Oxide-Semiconductor (CMOS) technology, the dynamic power dissipation ($P_{dynamic}$) is mathematically governed by the following equation [5]:

$$P_{dynamic} = \alpha \cdot C_L \cdot V_{DD}^2 \cdot f$$

Where:

α represents the switching activity factor (how often the gates toggle).

C_L represents the parasitic load capacitance of the physical silicon routing.

V_{DD} is the core supply voltage.

f is the operational clock frequency.

Because dynamic power is directly proportional to the clock frequency (f), microprocessors running at 1.5 GHz(1,500,000,000 Hz) dissipate immense amounts of thermal energy.

Because our developed design is built strictly from primitive digital logic gates on an FPGA, we were able to aggressively step down the system clock without losing any operational speed. The Moore FSM was strategically designed to operate on a highly divided clock signal, stepping down the FPGA's internal 50 MHz oscillator to a mere **1 Hz clock** (one tick per second). Physical traffic lights do not need to change in microseconds; human drivers react in seconds.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

This thesis project successfully conceptualized, mathematically designed, synthesized at the Register-Transfer Level, and rigorously simulated a highly adaptive, density-based intelligent traffic light controller. By deliberately stepping away from the inherent processing limitations and software vulnerabilities of traditional microprocessor-based architectures, this research definitively proved that replacing fixed sequential polling loops with a custom, parallel-processing hardware Priority Encoder fundamentally optimizes intersection throughput.

The structural integration of a Moore Finite State Machine ensured that all state transitions occurred with absolute physical safety. By mathematically mandating un-skippable yellow warning states and all-red clearance intervals at the flip-flop level, the mechanical design actively eliminates the risk of high-speed T-bone vehicular collisions during rapid light changes. Rigorous software simulations executed within the Xilinx Vivado EDA environment confirmed that the architecture successfully fulfills all of its core engineering objectives: it dynamically routes green lights based on real-time 8-bit sensor arrays, it instantly preempts civilian cross-traffic for emergency vehicles with zero computational latency, and it utilizes mathematical wait-timers to guarantee that minor residential side roads are never permanently gridlocked by heavily congested major highways.

Furthermore, the hardware design exhibited inherent physical fault tolerance, safely navigating edge cases such as stuck, noisy, or blinded sensors by allowing the sequential starvation timers to organically override broken combinatorial inputs. By achieving an astonishing, mathematically verified **79.1% reduction** in idle waiting time compared to traditional deterministic fixed-timer methodologies, this purely digital VLSI Verilog implementation offers a

highly robust, thermally power-efficient, and immensely scalable engineering solution to the modern crisis of urban traffic congestion.

6.2 Future Scope of Work

While the simulated VLSI architecture has proven to be highly capable, logic-safe, and theoretically sound, several exciting and highly advanced avenues remain for expanding the project’s real-world infrastructural impact in future iterations:

1. **Physical Silicon Hardware Deployment:** The immediate, logical next step involves taking the fully verified Verilog bitstream and flashing it onto a physical FPGA platform, such as the Xilinx Artix-7 or Spartan-7 development board. This will allow for real-world hardware-in-the-loop (HIL) interface testing using physical inductive loop sensors, infrared laser break-beams, localized acoustic sensors, and physical high-voltage LED arrays.
2. **V2X (Vehicle-to-Everything) IoT Communication Integration:** Instead of relying solely on localized acoustic siren telemetry, the architecture can be massively upgraded to support modern 5G IoT V2X protocols. Ambulances and fire engines equipped with cellular IoT transmitters could broadcast their exact GPS coordinates, velocity, and trajectory directly to the FPGA’s external communication module. This would allow the central logic controller to mathematically predict the emergency vehicle’s exact time of arrival and proactively begin clearing the intersection queue long before the ambulance is even physically visible or audible.
3. **Synchronized Metropolitan ”Green Waves”:** By integrating ultra-low-latency IoT communication modules (such as LoRaWAN or localized 5G nodes), multiple independent FPGA intersection controllers could share their localized density data continuously across an entire city grid. A centralized cloud dashboard could then mathematically synchronize the respective FSMs of consecutive intersections, creating massive, uninterrupted ”green waves.” This advanced synchronization would allow

heavy civilian traffic clusters—or critical emergency response convoys—to traverse an entire city avenue from end to end without hitting a single red light, vastly improving the overall logistical efficiency and safety of the city’s smart infrastructure.

In addition to these directions, a structured transition plan from simulation to deployment should include phased validation milestones, beginning with controlled campus-scale pilot intersections and progressively expanding to mixed-traffic urban corridors. Each phase should formally record queue clearance time, emergency preemption response delay, and lane fairness indices so that operational gains are quantified under practical conditions rather than assumed from laboratory traces alone. This evidence-driven rollout can significantly improve confidence among municipal stakeholders, transport authorities, and funding bodies.

Another valuable future extension is integration of adaptive policy profiles for time-of-day behavior. Morning commuter peaks, school-zone windows, and late-night low-density operation can each benefit from different threshold parameters while preserving the same core safety FSM. Such configurable policy layering would retain hardware determinism while enabling administrative flexibility for diverse traffic patterns across weekdays, weekends, and special public-event scenarios.

Finally, long-term maintainability can be strengthened through a standardized diagnostics interface that periodically reports controller state statistics, override counts, and fairness-trigger frequency. These telemetry indicators can support preventive maintenance and calibration planning, ensuring that the deployed FPGA-based controller remains efficient, transparent, and dependable over its operational lifetime.

To further improve long-term deployment reliability, a formal field-validation matrix can be introduced for each pilot corridor. This matrix should include scenario classes such as peak-hour arterial dominance, off-peak sparse flow, emergency burst intervals, sensor fault emulation windows, and manual override recovery cycles. For each class, predefined acceptance thresholds can be tracked for queue clearance time, phase transition compliance, fairness-trigger latency,

and post-event stabilization time. Such a matrix creates consistency between engineering evaluation and administrative reporting, because every pilot city can use the same technical checklist while preserving local threshold tuning. It also reduces subjectivity during approval decisions by converting qualitative observations into repeatable, timestamped performance evidence.

A complementary enhancement is staged parameter governance. In many municipal deployments, thresholds are changed informally to address short-term complaints, which can gradually erode controller stability. A governance process can prevent this by requiring versioned parameter sets, mandatory offline replay validation, and controlled activation windows before changes are applied on live intersections. Under this method, each threshold update is accompanied by a traceable rationale, expected impact, rollback option, and verification report. The practical benefit is twofold: engineering teams maintain deterministic behavior boundaries, and transport authorities gain an auditable record of why and when operational behavior changed. This governance layer is especially important when multiple intersections are managed by different field teams, because it ensures policy consistency without suppressing legitimate local optimization.

In addition, future work can include resilience-oriented redundancy planning at the sensing layer. While the developed architecture already tolerates transient disturbances through debouncing and FSM discipline, sustained sensor degradation remains a realistic field risk. A hybrid sensing strategy—combining IR, loop proxy estimation, and confidence-weighted vote logic—can reduce single-point sensing bias without increasing decision latency beyond acceptable limits. Importantly, this redundancy should remain subordinate to the same deterministic priority chain so that reliability improves without introducing non-deterministic arbitration behavior. With proper calibration, redundancy can reduce false starvation triggers, improve demand estimation accuracy in adverse weather, and maintain emergency detection confidence under partial hardware faults.

Another strategically important extension is cross-intersection coordination policy that preserves local autonomy. Instead of enforcing a centrally

dictated timing regime, neighboring controllers can exchange compact state summaries—such as outgoing platoon density class, expected arrival window, and active emergency corridor flags—through lightweight communication channels. Each controller can then adjust local thresholds within bounded safety constraints while preserving its own deterministic FSM operation. This approach avoids brittle global coupling and still delivers corridor-level smoothing benefits. It also aligns well with phased adoption: corridors with upgraded infrastructure can coordinate immediately, while legacy intersections continue operating independently until they are modernized.

From a deployment economics perspective, lifecycle costing should be evaluated not only by initial hardware expense but by cumulative operational savings. Reduced idling translates into fuel savings and lower emissions, but administrative savings also emerge from predictable maintenance windows, fewer reactive interventions, and faster incident diagnostics. A structured cost model can therefore combine direct operational metrics (e.g., mean delay reduction, emergency clearance improvements) with maintenance metrics (e.g., mean time to detect faults, mean time to restore service, calibration frequency). Presenting this combined model helps decision makers compare the controller against fixed-time systems using a total-value framework rather than upfront procurement cost alone.

Finally, institutional adoption can be accelerated by defining a standardized training and handover package for municipal engineering staff. Even robust FPGA systems can underperform if field operators are not trained to interpret telemetry, validate updates, and diagnose sensor drift correctly. A practical package should include scenario-driven operation guides, minimal diagnostic command sets, periodic competency refresh sessions, and escalation protocols for safety-critical anomalies. When human-process readiness is addressed together with hardware readiness, the controller can transition from successful academic prototype to dependable public infrastructure component with sustained performance over many years.

Beyond training, governance continuity can be improved by introducing a periodic cross-functional review forum involving control engineers, transport

planners, emergency-service representatives, and maintenance supervisors. Such a forum enables policy refinements to be evaluated against both technical feasibility and public-service impact before field activation. For instance, a threshold adjustment that improves commuter flow may still be rejected if it increases emergency-route variability in critical hospital corridors. Conversely, a policy that prioritizes emergency traffic aggressively can be moderated when fairness metrics indicate sustained disadvantage for specific feeder roads. Integrating these perspectives before deployment reduces policy oscillation and helps maintain balanced corridor behavior over time. In this way, the proposed architecture is supported not only by deterministic hardware logic but also by a disciplined institutional process that preserves safety, fairness, and transparency throughout lifecycle operation.

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Contribution to Program Outcomes (POs) and Program Specific Outcomes (PSOs)

The project contributes to the attainment of the following Program Outcomes:

PO1: Engineering Knowledge

Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

This project applies digital electronics, computing, and VLSI/FPGA engineering fundamentals to develop a starvation-free adaptive traffic controller. Mathematical logic design, HDL-based computation, and FSM modeling are used to solve the complex problem of dynamic intersection management.

PO2: Problem Analysis

Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

The work identifies key limitations of fixed-time traffic systems, formulates the engineering problem of congestion/emergency delay/starvation, and analyzes prior literature to derive substantiated design conclusions with sustainability focus (reduced idling and emissions).

PO3: Design/ Development of Solutions

Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

A creative FPGA-based controller is designed with emergency preemption, fairness timers, and Moore FSM safety transitions. The solution addresses

public safety, urban mobility, energy-aware operation, and scalable infrastructure needs.

PO4: Conduct investigations of complex problems

Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

The project conducts simulation-based investigations through structured test scenarios, traffic modeling, waveform interpretation, and comparative timing analysis, producing valid conclusions on efficiency and safety behavior.

PO5: Modern Tool Usage

Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

Modern EDA/IT tools (Xilinx Vivado, RTL simulation, waveform analyzers) are selected and applied for design, synthesis, verification, and modeling, while acknowledging hardware and scenario-bound limitations.

PO6: The Engineer and The World

Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

The project evaluates societal and environmental impacts by improving emergency response, reducing commuter delay, lowering idling emissions, and supporting safer intersection operation with sustainability-oriented outcomes.

PO7: Ethics

Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

The controller prioritizes life-critical emergency movement while preserving fairness for all lanes, reflecting ethical engineering practice, human-centric design values, and responsible safety compliance intent.

PO8: Individual and Collaborative Team work

Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

The project was executed through collaborative team contributions across architecture design, Verilog coding, testbench development, analysis, and documentation, with effective individual and group participation.

PO9: Communication

Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

Findings are communicated through technical documentation, block diagrams, truth tables, simulation waveforms, and structured report writing, enabling clear engineering communication to academic and technical audiences.

PO10: Project Management and Finance

Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments. The work follows phased planning (problem definition, design, verification, analysis, reporting) and considers cost-effective, scalable FPGA realization, reflecting engineering management and practical decision-making principles.

PO11: Life-Long Learning

Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

The project demonstrates self-driven learning in HDL design, adaptive control strategies, and emerging smart-infrastructure technologies, along with critical thinking for continuous technical improvement.

2. Program Specific Outcomes (PSOs)

PSO1: Apply the knowledge of domain-specific skill set for the design and analysis of components in VLSI and Embedded systems.

The project applies domain-specific knowledge in FPGA-based digital system design for the implementation and analysis of adaptive traffic-control modules including priority encoder logic, fairness timers, and Moore FSM sequencing, thereby satisfying PSO1.

3. Alignment with Sustainable Development Goals (SDGs)

The project supports the following Sustainable Development Goals:

SDG 3: Good Health and Well-Being

The project contributes to this goal by enabling faster and safer emergency-vehicle passage at intersections.

SDG 9: Industry, Innovation and Infrastructure

The project contributes to this goal by proposing an FPGA-based, modern, scalable, and intelligent traffic-control infrastructure.

SDG 11: Sustainable Cities and Communities

The project contributes to this goal by improving urban mobility, reducing congestion hotspots, and supporting safer road-network behavior.

SDG 13: Climate Action

The project contributes to this goal by reducing idling-related emissions through demand-aware signal allocation.

Overall Impact

The project not only fulfills academic learning outcomes but also contributes toward broader societal and environmental objectives. By integrating technical knowledge with sustainability considerations, the work demonstrates alignment between institutional Program Outcomes and global development priorities.

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